

Chapter 31 — Tuberculosis and Parasitic Infections of the Genitourinary Tract

UROGENITAL TUBERCULOSIS

- **Tuberculosis (TB) can involve any organ**, including the urogenital (UG) tract. Untreated UG TB causes **irreversible tissue destruction (renal failure, infertility)**, so belongs in the differential of UG disorders.
- **TB is the second "great imitator"** (after syphilis); mimicking other diseases delays diagnosis and treatment.
- **As TB becomes rarer in industrialised nations, diagnosis depends on deliberate clinical suspicion.**
- **Terminology:** "urogenital TB" is preferred over "genitourinary TB" because the **urinary tract is involved far more often than the genital tract.**

History

- **Mycobacterium tuberculosis has co-evolved with humans.** A progenitor (*M. prototuberculosis*) may have infected early hominids > 3 million years ago; a *Homo erectus* skeleton about 500,000 years old has TB-consistent bony lesions.
- **Oldest microbiological confirmation:** Mycobacterium tuberculosis complex DNA in **9000-year-old Neolithic skeletons** (Eastern Mediterranean). Tubercle bacilli shown microscopically and molecularly in **Egyptian mummies (c. 3000 BCE).**
- **Written descriptions:** ancient East Asia, pre-Columbian Americas, classical Greece and Rome, through modern history.
- **Epidemic peak: 18th and 19th centuries** in Europe and North America; "consumption" caused up to **one-quarter of deaths in the Industrial Age.**
- **Turning point, 24 March 1882: Robert Koch** presented the first isolation and identification of the tubercle bacillus; **24 March is World Tuberculosis Day.**

Microbiology

- **TB is caused by the Mycobacterium tuberculosis complex (MTBC):** closely related **acid-fast bacteria.**

MTBC SPECIES BY HOST RANGE

Host range	Species
Humans only	<i>M. tuberculosis</i> , <i>M. africanum</i>
Humans and other mammals	<i>M. bovis</i> , <i>M. canettii</i> , <i>M. microti</i> , <i>M. caprae</i> , <i>M. orygis</i> , <i>M. pinnipedii</i>
Animals only (so far)	<i>M. mungi</i> , <i>M. suricattae</i>

- **Commonest human isolate worldwide: *M. tuberculosis*, then *M. bovis*. Regionally, in West Africa *M. africanum*** causes a large share.
- **"*M. tuberculosis*" is often used, inaccurately, for the whole complex.**
- **The species are clinically indistinguishable but differ in drug susceptibility:** classically, *M. bovis* is innately resistant to pyrazinamide, a first-line agent.

- The table sorts species by host. The trap is assuming uniform susceptibility: **species matters for treatment, not clinical picture** — relevant to bacille Calmette–Guérin (BCG, derived from *M. bovis*) infection after intravesical therapy.

Concept — *the complex is one disease clinically but several organisms pharmacologically: identify the species because it changes the drug regimen, not because it changes the presentation.*

Epidemiology

- **Global (World Health Organization, WHO): one-quarter of the world's population has latent MTBC infection.** 2021: about **10.6 million new active cases** and **1.6 million deaths**. Deaths have fallen steadily since 2000; **TB mortality down 45% since 1990**.
- **Obstacles to control:** the **human immunodeficiency virus (HIV) epidemic in sub-Saharan Africa**, **global rise in obesity and diabetes**, and **multidrug and extensive drug resistance**.
- **United States:** about **8300 active cases in 2022 (2.5 per 100,000)**, declining steadily since the **1980s resurgence (peak 1992)**. Cases fell during COVID-19 (2020), back to pre-pandemic levels by 2022.
 - **Disproportionate burden in people born outside the United States:** incidence **16 times** that of US-born people.
- **UG involvement in pulmonary TB depends on setting:**

UROGENITAL TB BY SETTING

	Developing countries	Developed countries
Share of global UG TB	> 90%	< 10%
UG TB as a proportion of pulmonary TB	15–20%	2–10%
Rank among extrapulmonary sites	Second (after lymph nodes)	Third in the United States (after pleural and lymphatic TB); about one-quarter of extrapulmonary cases

- **Sex and age:** about **two-thirds male**; **mainly adults**, though reported in children as young as 2 years.

Concept — *UG TB burden tracks the TB burden, and diagnostic delay tracks resources. The developing world has more cases, a larger UG share and more advanced disease at presentation, and this later shows up in the rates of contracted bladder.*

Infection, Host Immune Response and Transmission

- **Entry is by inhalation of cough-generated aerosols** (direct soft-tissue inoculation is rare) → **alveolar macrophages phagocytose the bacilli**, with three possible outcomes:
 - **Cleared:** macrophages kill the bacilli; **no infection and no adaptive immune response**.
 - **Established:** bacilli escape killing and replicate in macrophages. **A cellular immune response may take up to 12 weeks to appear**; meanwhile bacilli spread **via lymphatics to the hilar nodes**, then **via the bloodstream to distant organs** — this is how the kidney and epididymis are seeded.
- **Granuloma formation is how the host contains infection:**
 - **Infected macrophages secrete cytokines:** interleukin-6 (IL-6), IL-12, IL-1 β , tumour necrosis factor- α (TNF- α).
 - **Structure:** a **centre of foamy macrophages, epithelioid cells and multinucleated (Langhans) giant cells** within a **lymphocyte cuff**.
 - **Antigen presentation activates T cells**, which secrete IL-2, TNF- α and, **most importantly, interferon- γ (IFN- γ)**, which maintains the granuloma and drives killing of infected macrophages and bacilli.

- If killing fails, the granuloma sequesters viable bacilli, which become dormant.

LATENT INFECTION VS ACTIVE DISEASE

	Latent TB infection	Active TB disease
Frequency after infection	90–95% controlled at this stage	< 5% progress within the first year (primary progression)
Pathology	Scarring (cicatrisation), granuloma calcification	Renewed replication → caseation (characteristic lesion) → bacillary release
Later risk	Lifetime reactivation risk 5–15%, higher with comorbidity	—
Infectious?	No	Pulmonary disease: yes (respiratory aerosols to close contacts)

- **Reactivation triggers (balance shifts towards the pathogen):** old age, renal failure, diabetes mellitus, malnutrition, HIV infection, other immunosuppression.
- **Isolated extrapulmonary (including UG) TB is generally not infectious**, except by **direct skin inoculation or aerosol-generating procedures on infected fluids or tissues** — relevant in theatre.
- **The latent/active split is a simplification.** TB is a **dynamic spectrum** — contained infection → subclinical disease → progressively infectious disease — and a person can **move in either direction** as host immunity (comorbidity) or the bacilli (treatment of latent infection) change.

Concept — TB outcome is a balance between host and pathogen, and the granuloma is where that balance is held. Latency, reactivation and the whole spectrum are the granuloma holding, failing or re-forming as host immunity changes, and IFN- γ is the cytokine that holds it.

Development of Urogenital Disease

ROUTES TO UROGENITAL TB

Route	Frequency	Mechanism and examples
1. Haematogenous	Principal route	Seeds kidneys and epididymis first; other UG organs then infected by contiguous spread from these sites. Seeding may be confined to the UG tract or widely disseminated
2. Ascending (retrograde)	Much less common	Classically intravesical BCG for bladder cancer (live, attenuated <i>M. bovis</i> vaccine): UG TB in about 1% (0.9%) of treated patients, as pyelonephritis, renal abscess, ureteric obstruction, cystitis, prostatitis or epididymo-orchitis
3. Contiguous spread	Rare	TB ignores anatomical boundaries: spine → psoas → kidney ; gastrointestinal (GI) TB → enterorenal and enterovesical fistulae
4. Direct inoculation	Exceedingly rare	Self-inoculation of external genitalia from infected stool or urine; person-to-person genital inoculation from infected genital or oral lesions

- **Haematogenous disease may appear soon after seeding or after latency of decades (up to 46 years)** — hence UG TB often presents long after, or without, pulmonary disease.

Concept — UG TB is usually a blood-borne disease that lands in the two best-perfused UG organs, the kidney and the epididymis, and spreads from there. The distribution of disease in the rest of the tract follows from those two starting points.

Clinical Manifestations and Pathological Features

- **Presentation is non-specific:** patients are often **repeatedly treated for bacterial infection or investigated for malignancy** first.

- **Symptoms depend on site and severity. Renal TB can be destructive yet silent until it reaches the bladder.**
- **Asymptomatic** in about 8% in developed countries (earlier presentation).
- **Constitutional symptoms (fever, weight loss, night sweats, malaise) in < 20%**; their absence does not exclude UG TB.
- **Lower urinary tract symptoms dominate: up to half dysuria alone, about half storage symptoms, about one-third haematuria and flank pain.**
- **Renal colic in < 10%**, from passage of **necrotic papillae, clot, stones or caseous debris** in severe pyelonephritis.
- **Hallmark laboratory finding: sterile pyuria ± haematuria**, in **> 90%** in developing countries.

UROGENITAL TB BY SITE — FREQUENCY, ROUTE AND SIGNATURE LESION

Site	Frequency in UG TB	Route	Signature lesion or sign
Kidney	80% (commonest)	Haematogenous	Caseating cavities → autonephrectomy (up to one-third)
Ureter	—	Descent from kidney	Distal stricture at the vesicoureteric junction; beaded corkscrew ureter
Bladder	—	Descent from kidney	Patchy cystitis → contracted "thimble" bladder (< 20 mL)
Epididymis	10–55% (second commonest); bilateral in about one-third	Haematogenous	Starts in globus minor (tail) ; posterior scrotal sinus in up to half
Vas deferens	—	Contiguous from epididymis	Thickened, beaded vas
Testis	Isolated disease rare	Contiguous from epididymis	Hard mass mimicking tumour ; hydrocele in about 5%
Prostate	22–49%	Haematogenous or urinary	Chronic prostatitis unresponsive to antibiotics
Seminal vesicles	—	Via vas or ejaculatory ducts	Infertility, possibly the first symptom
Urethra	1.9–4.5% (relatively resistant)	Usually with prostatic disease	Urethroscrotal fistula
Penis	Exceedingly rare	Primary, autoinoculation or haematogenous	Ulcerating papule or plaque → cavernosal nodules

- **What the table is for:** frequency follows route — **haematogenous landing sites (kidney, epididymis) commonest**, descent or contiguity next, **urethra and penis rare**.
- **The trap:** the **epididymis, not the prostate, is the commonest genital site**, and the urethra is relatively spared despite constant exposure to infected urine.

Kidney

- **Renal disease is progressive and highly destructive**, pathology varying with severity.
- **Kidney as collateral damage of pulmonary or systemic TB (no UG imaging changes):**
 - **TB granulomatous interstitial nephritis:** pulmonary TB + renal failure, ± pyuria, normal UG imaging; biopsy: granulomas, sometimes caseating. **Treatment can reverse the renal impairment.**
 - **Immune-complex glomerulonephritis** and **secondary amyloidosis**.

- **Miliary TB (widely disseminated):** heavy haematogenous load gives **innumerable pale granulomas about 3 mm across, like millet seeds**, studding cortex and medulla. **High mortality**, but the renal "milia" usually spare renal function.
- **Localised renal TB sequence:** bacilli lodge in **periglomerular capillaries** → parenchymal granulomas coalesce → **caseation forms cavities** → abscess, chronic pyelonephritis, **parenchymal and papillary necrosis** → **flank sinus tracts**.
 - Examination: **costovertebral angle tenderness**.
 - With advancing disease the **calyces become inflamed, then calcify**, causing calyceal **distortion, dilatation and stenosis**.
- **Autonephrectomy (non-functioning kidney)** has two types:

AUTONEPHRECTOMY — THE TWO TYPES

Caseo-cavernous	Fibrotic
Viable tissue replaced by granulomas and exudate-filled cavities	Severe scarring and calcification
± calcification	Shrunken kidney ("putty" kidney on plain film)

- **End-stage renal failure** in about 7%.
- **Squamous metaplasia of the renal pelvis** can **persist after treatment**, giving a **long-term risk of squamous cell carcinoma (SCC)**.

Ureter

- **Ureteric TB descends from the kidney:** urinary bacilli form **mural granulomas**, and **infected calculi can descend and impact**.
- **Inflammation** → **scarring** → **stricture, commonest distally at the vesicoureteric junction**; pan-ureteric strictures give a **"beaded corkscrew" ureter**.
- **A scarred, distorted ureter can both obstruct and reflux. Stricture-related obstruction is an important cause of renal failure in UG TB.**
- **Spontaneous ureteric perforation** above a vesicoureteric junction stricture (single case): stenting, medical treatment, reconstruction.

Bladder

- **Bladder TB descends, beginning around the ureteric orifices and spreading along lymphatics.** Urothelial implantation causes **patchy cystitis**; **ulcers** form where large granulomas coalesce.
- **The dome is most affected; trigone and bladder neck usually spared.** Mucosal inflammation, friability and haematuria follow.
- **After about a year of chronic inflammation and scarring, the bladder contracts:**
 - Capacity < 100 mL → **prominent frequency, urgency, pain and dysuria**.
 - **"Thimble" bladder:** capacity < 20 mL.
- **Contracture is late and commoner in the developing world** (about 12% vs 4% of UG cases) because diagnosis there is made later.
- **Rare: acute abdomen from spontaneous bladder perforation**, often recognised only at operation, before TB is suspected.

Epididymis, Vas Deferens, Testes and Scrotum

- **Second commonest site of haematogenous seeding, after the kidney.**

- **Disease begins in the more vascular globus minor:** epithelial granulomas → fibrosis and **luminal obliteration** → large caseous granulomas → **nodular epididymis, swollen or hard** on examination.
- **Granulomas can adhere to the skin and ulcerate**, forming the posterior scrotal sinus.
- **Vas deferens:** thickened and beaded from **nodular scarring**.
- **Testis: isolated epididymal or testicular TB is rare;** epididymal TB usually **extends into the testis**, granulomas replacing seminiferous epithelium and connective tissue with fibrosis to produce the tumour-like mass.

Prostate and Seminal Vesicles

- The prostate is reached by either route, and the route determines the picture:

PROSTATIC TB — HAEMATOGENOUS VS URINARY ROUTE

Haematogenous	Urinary
Peripheral lesions; urethra spared	Urethra involved
Asymptomatic → calcification, hard gland	Resembles bacterial prostatitis

- **Examination:** nodules or fluctuance.
- **Suspect TB in chronic prostatitis persisting despite antibiotics.** Quinolones are active against MTBC, but the **short courses for bacterial prostatitis are inadequate**, so symptoms persist or **recur quickly** — the partial response is the clue.
- **Prostatic abscess is rare**, particularly in **acquired immunodeficiency syndrome (AIDS)**.
- **Seminal vesicles:**
 - Routes: via the **vas** (from epididymal or testicular TB), or via the **urethra and ejaculatory ducts** (from renal, bladder or prostatic TB).
 - Pathology: mural granulomas → **luminal caseation** → **calcification**.
 - Features: **low-volume ejaculate, oligospermia, azoospermia, haemospermia**; rarely, abscess.
 - Examination: **enlarged vesicles early; hard nodules late**.

Urethra and Penis

- **Urethral TB usually accompanies prostatic TB;** isolated urethral TB is very rare.

PENILE TB — THE THREE FORMS

	Primary penile TB	Orificial TB	Papulonecrotic tuberculid (PNT)
Nature	True infection	True infection; rapidly necrotic	Hypersensitivity reaction to MTBC antigens carried to skin from a distant focus
Origin	Penile skin	Autoinoculation from own infected stool or urine ; rarely haematogenous or lymphatic	Disseminated antigen (commonest on glans; other skin sites possible)
Lesion	Inflamed papule or keratotic plaque (lupus vulgaris) → ulcer → spreads to corpora (pea-sized nodules , painless and hard, or fungating mass)	Painful ulcers with pseudomembrane , eroding deeply	Red papules → ulcers → varioliform scarring , recurrent
Pain	Painless nodules	Painful	Can be painless

	Primary penile TB	Orificial TB	Papulonecrotic tuberculid (PNT)
Bacilli, culture, polymerase chain reaction (PCR)	Present	Present	Absent: culture- and PCR-negative ; histology often inconclusive (mature granulomas often absent)
Host and prognosis	Fibrosis → penile distortion, erectile dysfunction	Immunocompromised or debilitated ; marks very advanced UG or GI TB ; poor prognosis	Reported from Japan, South Africa, India
Mimics	Malignancy	—	Syphilis, Behçet disease, recurrent herpes simplex, balanitis, SCC
Diagnosis	Tissue	—	Recognition + response to empirical anti-TB treatment despite negative cultures

- **What the table tests:** in PNT alone **negative cultures and PCR do not exclude TB**; **response to treatment is the diagnostic test**.

Concept — In UG TB the route of spread predicts the lesion. Haematogenous seeding gives parenchymal disease (kidney, epididymis, peripheral prostate). Descending urinary spread gives luminal and orifice disease (distal ureteric stricture, cystitis, urethral prostatic disease). Healing by fibrosis turns an infection into an obstruction (stricture, contracted bladder, obliterated epididymis, infertility).

Diagnosis

- In developed countries the goal is MTBC culture for drug-susceptibility testing. Order of evidence:
 - Culture or DNA positive → **diagnosis**.
 - Negative → **caseating granulomas on tissue** in the right clinical context support TB.
 - Also absent → **compatible clinical picture + probable exposure + response to empirical treatment**.
- Up to one-fifth of UG TB coexists with pulmonary TB: always look for pulmonary disease.

Culture

- **Gold standard: urine acid-fast bacilli (AFB) culture.**
- **Sampling: first-void (most concentrated) urine, 3–5 samples on consecutive days, cultured immediately** because prolonged exposure to acidic urine inhibits mycobacterial growth. **Culture any biopsy or operative tissue too.**
- **Sensitivity depends on technique and shedding:** up to **80–90%** with **ideal technique**; as low as **10%** in **real-world practice**, compounded by **intermittent bacillary shedding**.
- **Ziehl–Neelsen stain of urine:** sensitivity < **50%**.

MYCOBACTERIAL CULTURE SYSTEMS

System	Medium	Time to detection	Setting and notes
Löwenstein–Jensen (LJ)	Solid, egg-based	4–6 weeks	Medium of choice in developing countries: cheapest, no special equipment; laborious
Middlebrook 7H10	Solid, agar-based, transparent	About 1 week sooner than LJ	Developed countries; costlier

System	Medium	Time to detection	Setting and notes
BACTEC Mycobacteria Growth Indicator Tube (MGIT)	Liquid , fully automated; fluorescence quenching	As little as 10 days	Developed countries; first-line susceptibility testing done in-house on MGIT
Semi-automated radiometric	Liquid	—	Alternative

- **Rule:** culture on at least one solid medium alongside the liquid system to maximise yield.
- **Susceptibility testing:** any culture method can be used; **second-line drugs and molecular resistance sequencing are done in reference laboratories.**

Nucleic Acid Amplification Tests

- **Nucleic acid amplification tests (NAATs)** give results in **1–2 days** and detect MTBC when **bacillary load is too low** for culture.
- **Natural amplification inhibitors in urine** lower sensitivity in non-respiratory specimens; sensitivity also varies with specimen, target sequence and protocol.
- **GeneXpert MTB/RIF (marketed since 2010):** self-contained, automated (sputum processing → DNA extraction → amplification); **detects MTBC and rifampicin resistance simultaneously.**
 - Rifampicin resistance is a surrogate for multidrug-resistant TB (MDR-TB), as **> 90% of rifampicin-resistant strains are also isoniazid (INH)-resistant.**
 - MDR-TB = resistance to both INH and rifampicin.
- **Performance in urine:** systematic reviews show **good sensitivity and specificity for GeneXpert against urine AFB culture**, but small, heterogeneous studies make estimates imprecise. NAAT on **epididymal biopsy tissue** performed well in a single study.
- **Role of NAAT:**
 - **United States:** adjunct only; it **does not replace culture**, and lacking high-quality sensitivity data, a **negative NAAT does not exclude extrapulmonary TB.**
 - **WHO:** GeneXpert MTB/RIF is **part of the initial work-up of all patients with TB**; **cost and equipment** are the barriers in developing countries.
 - **NAATs cannot monitor treatment response:** **dead organisms still shed nucleic acid**, so the test stays positive despite adequate treatment. **Culture is used for monitoring.**

Urine Lipoarabinomannan Assay

- **Lipoarabinomannan (LAM)** is a **mycobacterial cell-wall glycolipid** found in urine, sputum and serum. Methods range from enzyme-linked immunosorbent assay (ELISA) to the **point-of-care lateral-flow urine assay (LF-LAM): no laboratory equipment, result in < 30 minutes.**
- **In people living with HIV (PLHIV):** a systematic review showed **low sensitivity, high specificity;** sensitivity somewhat better with TB symptoms. **A positive result is useful; a negative result is not reassuring.**
- **WHO (2019) recommends adding LF-LAM for:**
 - PLHIV with TB symptoms
 - People with advanced HIV disease who are unwell
 - PLHIV with a CD4 count **< 200 cells/mm³**
- **United States:** broad access to other diagnostics confines LF-LAM mainly to academic centres.

Histopathology

- **Biopsy when urine cultures are negative.** Mycobacteria are often **not seen in tissue**, but **caseating granulomas in the right clinical context** support the diagnosis.
- **Histology is consistent with TB in only a minority** (about one-third in developed countries, one-fifth globally). **A negative biopsy does not exclude TB.**
- **Epididymal nodules: fine-needle aspiration cytology diagnostic in about two-thirds.**
- **Adding a TB NAAT to tissue increases sensitivity.**

Screening Tests

- **The tuberculin skin test (TST) and interferon- γ release assays (IGRAs) cannot distinguish latent infection from active disease.**
 - Both are United States Food and Drug Administration (FDA)-approved and widely used but of **limited value for active disease: a positive test does not rule TB disease in; a negative test does not rule it out.**
 - **Their proper use is screening for TB infection.**
- **Preferred screening test: IGRA for persons ≥ 5 years born in TB-endemic countries; TST remains acceptable.**

Tuberculin Skin Test (Purified Protein Derivative, Mantoux Test)

- **Principle:** detects an **existing cellular (delayed-type hypersensitivity) response** to MTBC antigens.
- **Technique:** tuberculin (sterile protein extract of *M. tuberculosis*) injected **intradermally into the volar forearm; induration read at 48–72 hours.**
- **Centers for Disease Control and Prevention (CDC) cut-offs are risk-stratified:**

TST — POSITIVITY THRESHOLDS BY RISK GROUP (CDC)

Induration	Positive in
≥ 5 mm	Recent contact with TB; fibrotic chest radiograph changes consistent with old TB; immunosuppressed
≥ 10 mm	Immigrants from high-prevalence countries; residents of or workers in high-risk institutions; people who inject drugs; comorbidities raising active-TB risk
≥ 15 mm	General population with no risk factors

- **The logic of the table: higher pre-test probability, lower cut-off** — a small reaction matters in someone likely infected or unable to respond. The trap is one cut-off for everyone.
- **TST advantages: cheap, no laboratory, easy** (though staff must be trained to place and read it).
- **TST disadvantages:**
 - **Not MTBC-specific: BCG vaccination and non-tuberculous mycobacteria** can cause positives.
 - **Second visit** needed for reading.
 - **False negatives in 10–25% of active TB.**
- **In UG TB the TST is usually positive.**

Interferon- γ Release Assays

- **Principle:** blood test measuring IFN- γ released by sensitised T cells to MTBC-specific antigens — an **in vitro, MTBC-specific TST.**
- **Its antigens are absent from all BCG strains and most non-tuberculous mycobacteria, so no BCG false positive** — the key advantage over TST.

TST VS IGRA

	TST	IGRA
Format	Intradermal, in vivo	Blood, in vitro
Visits	Two (read at 48–72 h)	One (result about 24 h)
Affected by BCG?	Yes: false positive	No
Affected by non-tuberculous mycobacteria?	Yes	Mostly no
Laboratory	Not needed	Needed
Latent vs active	Cannot distinguish	Cannot distinguish

THE TWO IGRAS AVAILABLE IN THE UNITED STATES

	QuantIFERON-TB Gold Plus (QFT-Plus, fourth generation, 2017)	T-SPOT.TB
Sample	Whole blood in four tubes: ESAT-6/CFP-10; proprietary antigens that activate CD8 T cells ; negative control; positive control	Separated peripheral blood mononuclear cells (PBMCs) in antibody-coated wells with ESAT-6 and CFP-10
Incubation	16–24 h in collection tubes	—
Read-out	Plasma IFN-γ concentration by ELISA	Enzyme-linked immunospot (ELISPOT): spots (IFN-γ-secreting cells) counted manually
Results	Positive, negative or indeterminate	Positive, negative or borderline
Ease	Easier	Harder to perform
Sensitivity	Third-generation QFT least sensitive of the three tests	Most sensitive (T-SPOT.TB > TST > third-generation QFT in culture-confirmed TB)

- Pooled sensitivity data for fourth-generation QFT-Plus are **not yet available**.

Concept — Culture proves disease; NAAT and LAM speed it up; TST and IGRA only prove exposure. Every test in this section is placed by what it can and cannot do: only culture gives susceptibility and tracks treatment, and only immune-response tests are blind to the difference between latent and active TB.

Radiography

- **Imaging often first suggests TB in a UG disorder. Choose modality by site and clinical picture.**

Plain Radiography (Kidney–Ureter–Bladder Film)

- **Calcification is frequent**, in about **a quarter to two-fifths** (24–44%), evolving with disease:
 - **Early:** faint punctate parenchymal calcifications.
 - **Progressive:** **globular calcification** of a tuberculous mass (Fig. 31.1).
 - **Papillary necrosis:** **triangular, ring-like calcifications** in the collecting system.
 - **Fibrotic autonephrectomy:** small, shrunken, calcified "**cement**" or "**putty**" kidney with calcific rims outlining individual lobes — **this lobar pattern is pathognomonic of end-stage renal TB.**
- **Infected stones** take odd shapes in a deformed, fibrosed pelvis; an **upward-pointing arrowhead stone** suggests a pelvis "**hiked up**" by scar contraction.
- **Ureteric calcification:**

URETERIC CALCIFICATION — TB VS SCHISTOSOMIASIS

TB	Schistosomiasis
Intraluminal	Mural
• Bladder wall calcification is uncommon except late, with contracted bladder. Prostatic and seminal vesicle calcification in about 10%. • Beyond the urinary tract: vertebral body erosions, calcified psoas cold abscesses. • Every patient evaluated for UG TB needs a chest radiograph to exclude coexistent infectious pulmonary TB.	

FIG. 31.1 Kidney-ureter-bladder film: left renal TB with associated calcification.

Intravenous Urography

- **Intravenous urography (IVU), the former gold standard for early renal TB**, gave most classic signs. Still called the preferred first test in some regions, it is **essentially replaced by computed tomography (CT), CT urography, magnetic resonance imaging (MRI) and ultrasound (US)**.
- **Early signs: loss of sharpness, irregular urothelial margins; "moth-eaten" calyces.**
- **Filling defects from a tuberculoma rupturing into a calyx or papillary necrosis; medullary cavities communicating with the collecting system.**
- **"Phantom calyx":** a stenosed calyx or infundibulum excretes no contrast, so the calyx is **absent where it should be**.
- **Ureter:**
 - **Pipestem ureter:** rigid, calcified, straight, tubular, **aperistaltic**.
 - **Beaded corkscrew ureter:** nodular fibrosis along the whole ureter.
 - **Both highly suggestive of TB, especially with kidney or bladder changes.**
- **Non-functioning kidney (autonephrectomy); fibrosed, contracted bladder** (Fig. 31.2).
- **Perinephric abscess** suggested by **loss of renal respiratory movement or ureteric displacement**.
- **Commonest IVU findings are obstructive changes from scarring**, not the classic erosive signs:
 - **Calyceal obliteration, infundibular narrowing, hydrocalycosis, segmental or total hydronephrosis, hydroureter** (Figs. 31.3, 31.4)
 - **Calyceal dilatation and distortion → "cloverleaf" pattern**
 - **Vesicoureteric junction obstruction** from tuberculous cystitis or **distal-third ureteric stricture** (Fig. 31.5)
 - **Kerr's kink: "hiked-up" renal pelvis with sharp pelviureteric junction angulation**

FIG. 31.2 Cystogram phase of an intravenous pyelogram: left renal TB, with **contraction of the left side of the bladder** caused by tuberculous fibrosis. FIG. 31.3 Occluded calyx. FIG. 31.4 Severe calyceal and parenchymal destruction. FIG. 31.5 Stricture of the distal left ureter.

Computed Tomography With Urography

- **CT urography, the most used modality in developed countries, has replaced IVU there**, showing KUB and IVU features (calcification, scarring, obstruction) **more sensitively**:
 - **More sensitive than KUB for calcification and collecting-system wall thickening.**
 - Multidetector scanners detect **3–4 mm lesions**.
 - **Intravenous contrast assesses function** across excretory phases.
- **Granulomas:**
 - **Small: hypo-attenuating, little or no enhancement.**

- **Large: mass-like, peripherally enhancing**, can **mimic renal cell carcinoma**.
- **Particularly useful in complicated or extensive disease:** perinephric and psoas abscesses; lymph nodes, vertebrae, spleen, liver; **prostate and seminal vesicles** (enlargement, necrosis, cavitation, abscess, calcification).
- **CT vs IVU:**
 - **CT disadvantages:** **higher radiation dose; less sensitive for earliest changes** (minimal urothelial thickening, subtle papillary necrosis), **for which IVU may still be preferred in resource-limited settings**.
 - **Contrast CT advantages:** shows **parenchyma, tissue oedema and hypoperfusion**.

FIG. 31.6 CT with oral contrast in bilateral TB: the **right kidney is hydronephrotic from infundibular stenosis but retains good function**; the **left kidney is an end-stage, non-functioning, atrophic, calcified kidney**. The figure illustrates the **two end-points side by side: obstruction that can be salvaged vs autonephrectomy**.

Retrograde and Antegrade Pyelography

- **Both replaced by CT urography;** useful **when IVU or CT is impossible** (renal insufficiency, contrast allergy) to delineate distorted anatomy.
- **With IVU, they show whether cavities are obstructive and whether they communicate with the collecting system.**

Ultrasonography

- **Limited diagnostic role:** findings **non-specific**, visualisation poorer than CT.
- **No radiation and low cost** make it the **modality for children and pregnancy**.
- **Uses:**
 - **Scrotum:** diffusely enlarged testis or epididymis with **homogeneous or heterogeneous hypoechoic areas, complex collections, hydrocele**.
 - **Transrectal US:** **thickened prostate and seminal vesicles**.
 - **Kidney:** abscesses, cavities; **septated cystic lesions suggest chronic infection**; calcifications **echogenic with distal acoustic shadowing**.
 - **Restricted renal respiratory movement suggests perinephric or psoas abscess**.
 - **Abdomen:** **ascites, lymphadenopathy, omental caking**.
- **Main use:** following hydronephrosis during medical treatment, because **healing fibrosis can worsen obstruction** while the infection is cured.

Magnetic Resonance Imaging

- **MRI is used less than contrast CT;** radiation-free, so **useful in children and pregnancy**. Findings resemble CT, but **better tissue resolution** can show a **single granuloma**.
- **Signal characteristics:**
 - **Small lesions:** **hypointense on T1 and T2**.
 - **Larger lesions:** **centrally T2-hyperintense** from central cellularity.
 - **Large lesions can mimic malignancy**, not always distinguishable.
- **Magnetic resonance urogram (MRU):** more sensitive than IVU for urothelial thickening and caliectasis.
- **Diffusion-weighted imaging (DWI):**
 - Distinguishes **hydronephrosis from pyonephrosis**.

- **Monitors renal fibrosis: apparent diffusion coefficient falls as fibrosis increases**, allowing staging and assessment of treatment effect.
- **Cine-MRU and dynamic MRU** assess **ureteric peristalsis**.
- **Caution with gadolinium in renal failure: risk of nephrogenic systemic fibrosis.**

IMAGING IN UROGENITAL TB — MODALITY, ROLE AND SIGNATURE SIGNS

Modality	Current role	Signature signs and specific use
KUB	Frequently shows calcification	Lobar "putty" kidney (pathognomonic of end stage); intraluminal ureteric calcification; arrowhead stone
Chest radiograph	Mandatory in all patients	Coexistent pulmonary TB
IVU	Former gold standard ; still used in resource-limited settings for early disease	Moth-eaten calyx, phantom calyx , pipestem and corkscrew ureter, cloverleaf calyces, Kerr's kink
CT urography	Most used in developed countries	Calcification (more sensitive than KUB), granuloma mimicking renal cell carcinoma, abscesses, disease beyond urinary tract
Retrograde or antegrade pyelography	When contrast CT or IVU contraindicated	Whether cavities communicate with collecting system and are obstructive
US	Children and pregnancy; scrotum; transrectal	Serial monitoring of hydronephrosis during treatment
MRI / MRU with DWI	Children and pregnancy; lesion characterisation	Single granuloma; DWI: pyonephrosis vs hydronephrosis, grading of fibrosis

Cystoscopy and Ureteroscopy

- **Endoscopy has a limited diagnostic role: direct vision, but non-specific findings:**
 - Hyperaemia, erosion, ulceration, granulomatous masses, irregular ureteric orifices.
 - **Ulcers can mimic malignancy.**
- **A "golf-hole" ureteric orifice suggests TB → image or scope the upper tract.**
- **Biopsy when possible, especially if malignancy is possible.**
- **Why biopsy though urine culture is diagnostic: culture may be too slow, and in culture-negative patients bladder biopsy has moderate yield (19–52% sensitivity).**

FIG. 31.7 Endoscopic and gross appearances of UG TB:

- (A) Extensive TB of the kidney and ureter with calcification and stricture
- (B) Acutely inflamed ureteric orifice
- (C) Tuberculous bullous granulations
- (D) Acute tuberculous ulcer
- (E) Tuberculous **golf-hole** ureteric orifice
- (F) Golf-hole orifice, **severely withdrawn** (retracted by ureteric fibrosis)
- (G) Healed tuberculous lesion
- (H) Acute tuberculous cystitis with ulceration

Concept — The imaging signs of UG TB are the pathology of fibrosis made visible. Destruction (moth-eaten calyx, cavities, calcification) is followed by healing that scars and shortens: phantom calyx, Kerr's

kink, pipestem ureter, golf-hole orifice and thimble bladder. That is why the commonest finding is obstruction, and why hydronephrosis must be watched during treatment.

Treatment

- **Before antimicrobials:** sanatorium rest and nutrition; for severe UG disease, **extirpative surgery was the only hope of cure.**
- **Drug milestones:** streptomycin 1944 → isoniazid (INH) 1952 → rifamycins 1957, replacing both sanatoria and surgery.
- **Today most patients, including MDR-TB, are treated medically as outpatients. Surgery establishes the diagnosis or is adjunctive in advanced disease.**

Medical Therapy

- **Multidrug therapy is mandatory, for five reasons:**
 - Bacilli live in **different micro-environments** with different metabolic states and replication rates.
 - Some drugs are **bactericidal**, others only **bacteriostatic**.
 - Some act best on **rapidly replicating** bacilli, others on **metabolically quiescent** ones.
 - Drugs differ in **tissue penetration and optimal pH**.
 - **Combination therapy prevents emergence of resistance.**
- **Start with four drugs:** INH, rifampicin, pyrazinamide and ethambutol (RIPE); first-line combination gives the **best cure rate in the shortest time.**
- **Baseline tests:** platelet count, liver and renal function; HIV test; where appropriate, **hepatitis B and C and diabetes screening.**
- **Tailor regimen to susceptibility. Directly observed therapy (DOT)** secures adherence and prevents resistance.
- **Second-line agents:** for intolerance, resistance or failure of first-line treatment; variable tolerability and ease of administration.
 - **Recent additions:**
 - **Fluoroquinolones and linezolid**, originally developed for other bacterial infections.
 - **Bedaquiline**, the **first new drug in 40 years developed specifically for TB.**
 - **Pretomanid** (FDA-approved 2019), **with bedaquiline and linezolid** for MDR-TB or extensively drug-resistant TB (XDR-TB).

TABLE 31.2 FIRST-LINE ANTITUBERCULOUS DRUGS

Drug	Adult daily dose*	Adult intermittent dose (3 times/week)†	Main adverse effects	Key notes
Isoniazid‡ (oral; injection available)	5 mg/kg (typically 300 mg)	15 mg/kg (typically 900 mg)	Hepatotoxicity, peripheral neuropathy	Give pyridoxine to prevent neuropathy
Rifampicin§ (oral; injection available)	10 mg/kg (typically 600 mg)	10 mg/kg (typically 600 mg)	Hepatotoxicity, flu-like syndrome, pruritus, drug interactions	Usually not combinable with protease inhibitors or certain NNRTIs in HIV
Rifabutin¶	5 mg/kg (typically 300 mg)	Not recommended	Hepatotoxicity, flu-like syndrome, uveitis, neutropenia, drug interactions	Rifamycin of choice with antiretrovirals (dose adjusted)

Drug	Adult daily dose*	Adult intermittent dose (3 times/week)†	Main adverse effects	Key notes
Rifapentine**	—	Once weekly , 10–20 mg/kg ($\leq$ 600 mg), continuation phase only, very selected patients	Similar to rifampicin	Contraindicated in HIV, cavitary pulmonary TB and extrapulmonary TB
Pyrazinamide	Weight-banded: 40–55 kg 1000 mg; 56–75 kg 1500 mg; 76–90 kg 2000 mg	40–55 kg 1500 mg; 56–75 kg 2500 mg; 76–90 kg 3000 mg	Arthralgia, hyperuricaemia, hepatotoxicity, pruritus, rash, GI upset	<i>M. bovis</i> innately resistant
Ethambutol	Weight-banded: 40–55 kg 800 mg; 56–75 kg 1200 mg; 76–90 kg 1600 mg	40–55 kg 1200 mg; 56–75 kg 2000 mg; 76–90 kg 2400 mg	Reduced red–green colour discrimination, reduced visual acuity, optic neuritis	Only first-line drug that is not hepatotoxic

*Or 5 times/week by DOT.

†**Daily therapy is preferred.** Intermittent therapy only in the continuation phase and only by DOT. **WHO and CDC no longer recommend dosing less than 3 times/week.**

‡**Pyridoxine 25–50 mg** if malnourished, pregnant, or with HIV, renal failure, thyroid disease, alcoholism or diabetes.

§Generally cannot be taken with protease inhibitors or certain non-nucleoside reverse transcriptase inhibitors (NNRTIs).

¶**Rifabutin dose depends on the antiretroviral:**

- **Efavirenz:** increase to **450 mg/day** (or 600 mg 3 times/week).
- **Fosamprenavir, nelfinavir or indinavir:** **150 mg/day** (or 300 mg 3 times/week).
- **Lopinavir/ritonavir, ritonavir or atazanavir alone, or ritonavir-boosted protease inhibitors:** **150 mg on alternate days** (or 3 times/week). Some experts consider this subtherapeutic and give 150 mg daily (or 300 mg 3 times/week) with close **uveitis** monitoring.

**Once-weekly rifapentine is not advocated by the WHO.

Source: American Thoracic Society, CDC and Infectious Diseases Society of America treatment guidelines (Medical Letter, 2012); American Thoracic Society, MMWR 2003.

- **What the table is for:** pairing each first-line drug with its signature toxicity.
 - **INH** → **neuropathy** (hence pyridoxine)
 - **Rifamycins** → **interactions** (hence rifabutin in HIV)
 - **Pyrazinamide** → **gout and arthralgia**
 - **Ethambutol** → **the eye**
- **The pattern:** every first-line drug except ethambutol is hepatotoxic; ethambutol alone needs visual rather than hepatic monitoring.
- **The trap:** rifapentine is listed but **contraindicated in extrapulmonary (therefore UG) TB**.

FIRST-LINE ANTITUBERCULOUS DRUGS — SIGNATURE TOXICITY AND MONITORING

Drug	Signature toxicity	Monitor or prevent
INH	Peripheral neuropathy; hepatitis	Pyridoxine; liver enzymes

Drug	Signature toxicity	Monitor or prevent
Rifampicin / rifabutin	Drug interactions; hepatitis; (rifabutin) uveitis	Antiretroviral and immunosuppressant levels; liver enzymes
Pyrazinamide	Hyperuricaemia, arthralgia; hepatitis	Liver enzymes
Ethambutol	Optic neuritis (red–green colour loss)	Visual acuity and colour vision

• **Standard short course for uncomplicated UG TB: 6 months:**

- **Intensive phase, 2 months:** daily INH + rifampicin + pyrazinamide, plus ethambutol until susceptibility is known; stop ethambutol if susceptible to the other first-line drugs.
- **Continuation phase, 4 months:** INH + rifampicin, daily (or three times weekly); twice-weekly dosing no longer recommended.
- **Pyridoxine (vitamin B6)** throughout against INH neuropathy.

TREATMENT PHASES — PURPOSE

Intensive phase (2 months)	Continuation phase (4 months)
Kills rapidly multiplying bacilli	Eradicates slow, sporadic multipliers and persisters
3–4 drugs	2 drugs

- **First-line drugs suit UG TB:** high urinary concentrations and good activity in acidic conditions.
- **The new 4-month regimen** (daily rifapentine, INH, pyrazinamide, moxifloxacin) for drug-susceptible TB is **not recommended for extrapulmonary TB and must not be used for UG TB.**
- **Prolonged treatment** (duration depends on disease and drugs):
 - **Complicated UG TB** (abscesses, severe scarring, extensive disease) and UG TB's **higher relapse rate than pulmonary TB** lead some to treat **12–24 months**, or **≥ 6 months after surgery.**
 - **Discuss any departure from the standard short course with a TB specialist.**
- **Monitoring during therapy:**
 - **Liver enzymes monthly if pre-existing liver disease.** Drug hepatotoxicity is reversible on stopping the drug; avoid alcohol and other hepatotoxic drugs. Severe liver injury has occurred.
 - **Renal lesions can worsen during treatment: healing fibrosis can aggravate obstruction and bladder contraction.** Steroids may help; surgery may be needed for new or worsening obstruction.

Concept — TB chemotherapy is designed around the biology of the bacillus. Mixed populations and different metabolic states explain why several drugs are used and why there are two phases. Resistance explains the four-drug start and DOT. The urinary tract adds one specific hazard: **cure comes through fibrosis, and fibrosis obstructs.**

Corticosteroids

- **Adjunctive steroids:** role not fully established. Rationale: limit excessive immune-mediated destruction and scarring.
- **Uses:**
 - **TB meningitis:** mortality benefit; strongly recommended.
 - **Selected cases:** severe pulmonary TB with paradoxical worsening on treatment; TB pericarditis (to prevent constriction).
 - **UG TB:** to prevent ureteric stricture and bladder contraction, on anecdotal evidence only; no trials.
- **A meta-analysis of trials in pulmonary, meningeal, pleural, pericardial and peritoneal TB showed reduced mortality whichever organ was involved.**

Surgical Therapy

- **About half (55%) of UG TB patients need surgery**, rising with stage.
- **Indications:**
 - Relieve obstruction and drain infected material
 - Remove a **non-functioning, infected kidney** that resists cure
 - **Medically resistant hypertension from a functionally excluded kidney**
 - **Reconstruct** the urinary tract
 - Abscesses unresponsive to medical treatment → **aspiration, or percutaneous or open drainage**
- **More than half of TB operations are now reconstructive.**
- **Optimal timing: 4–6 weeks after starting medical therapy**, letting **inflammation subside, bacillary load fall and lesions stabilise.**

Procedures to Relieve Obstruction

- **Uraemia or sepsis → emergency decompression.** Renal failure usually reflects **bilateral obstruction or obstruction of a functionally solitary kidney.**
- **Early stenting or percutaneous nephrostomy (PCN)** for tuberculous ureteric stricture **limits loss of function and preserves the chance of later reconstruction.**
- **Escalation:**
 - **Retrograde double-J stent (preferred)** → succeeds in **under half (41%).**
 - Fails → **antegrade stent**, internalised or externalised, via percutaneous puncture.
 - Fails → **PCN until definitive treatment**; loculation from strictures and scarring may need more than one.
- **PCN must be followed by correcting the cause**; removal without correction risks a **tuberculous cutaneous fistula** (less likely with effective concurrent chemotherapy).
- **Unsalvageable kidney → nephrectomy.**
- **Avoid high contrast-injection pressures** during stent or PCN placement, which could **disseminate infection.**

Nephrectomy

- **The overriding surgical goal is organ preservation.**
- **Partial nephrectomy: localised lesion (e.g. polar mass) unresponsive to medical therapy.**

TOTAL NEPHRECTOMY — INDICATIONS

Established indications	Contested indication
Non-functioning kidney + TB recalcitrant or recurring despite optimal therapy. Relapse after nephrectomy with short-course therapy < 1%	Non-functioning kidney without recalcitrant disease or hypertension. Some advocate removal, but no supporting evidence ; studies showing bacilli in removed kidneys predate modern chemotherapy
Non-functioning kidney + medically resistant hypertension. Hypertension improves in about two-thirds (65%)	

- **Nephrectomy in about one-quarter (27%) of UG TB patients**, similar in developed and developing countries.
- **Open technique:**
 - **Oblique retroperitoneal incision**, extendable dorsally or ventrally, because of dense fibrosis.

- **Remove granulomatous or caseous perinephric tissue with the specimen.**
- **Ligate renal artery and vein separately** to avoid a late arteriovenous fistula.
- **The ureter is usually left.**
- Minimise lymphatic disruption; **avoid entering pleura and peritoneum.**
- **Laparoscopic nephrectomy increasingly preferred** despite early fibrosis concerns: **less blood loss, faster recovery**; in experienced hands only **modestly longer** (about half an hour) than other indications.

Ureteropelvic and Ureteral Surgery

- **Temporarily stent a pelviureteric junction (PUJ) or ureteric stricture** to recover renal function before definitive repair.
- **Stricture site:**
 - **Upper and mid-ureteric strictures are rare**; may suit endourology.
 - **Lower ureteric strictures are common**; often need open surgery.
- **Decision factors:** stricture length and severity, guidewire passage, blood supply, renal function.

Endoscopic management

- **TB strictures have mucosal ischaemia and dense fibrosis**, so success rates for other strictures **do not transfer**.
- **Best results:** short stricture, residual lumen, good renal function.
- **Strictures forming during chemotherapy and stented early (double-J) can stabilise without further treatment.**
- **Balloon dilatation** (retrograde or antegrade): ureter, PUJ, vesicoureteric junction, calyceal infundibula; stent usually left. **High failure rates; repeat procedures common.**
- **Every ureteric stricture needs follow-up imaging (US or IVU)**, especially after endoscopic treatment, because **strictures can worsen as they heal.**
 - **Deterioration** → consider adding **corticosteroids**.
 - **No improvement or progression after 6 weeks of chemotherapy** → open surgery.

Open surgical options

- **Long, complex strictures need open repair**; mobilisation is hampered by **fibrosis, loss of elasticity and poor vascularity**.
- **PUJ repair in TB is harder than for congenital PUJ obstruction.**

OPEN RECONSTRUCTION BY STRICTURE SITE

Site	Procedure	Key point
PUJ, short segment, extrarenal pelvis	Dismembered pyeloplasty	—
PUJ, longer stricture	Non-dismembered (flap) pyeloplasty	May be impossible if pelvis heavily scarred
PUJ, reconstruction impossible	Ureterocalicostomy (ureter to lower-pole calyx)	Preserve renal capsule to cover the lower pole ; omentum if capsule insufficient, to prevent anastomotic stenosis
Upper or mid ureter	Excision + tension-free ureteroureterostomy; or Davis intubated ureterotomy (adhesiolysis + intubation)	—

Site	Procedure	Key point
Lower ureter	Complete excision back to healthy, well-vascularised ureter → ureteroneocystostomy	—
Small gap	Contralateral bladder mobilisation ± division of the superior vesical artery	Gains 2–3 cm
Moderate gap	Psoas hitch (needs good bladder capacity)	Bridges up to 5 cm; avoid genitofemoral and femoral nerves
Long gap	Boari flap ± psoas hitch	Bridges 10–15 cm; a poorly executed flap reduces bladder capacity; contracted TB bladder may be unsuitable
Multiple or recurrent strictures	Ileal ureteric replacement (interposition)	When native ureter no longer a usable conduit

- **The logic of the table:** gap length determines the manoeuvre (2–3 cm → 5 cm → 10–15 cm → ileum).
- **The trap:** in TB the bladder itself may be contracted, ruling out bladder-based bridges (psoas hitch, Boari flap).

Bladder Surgery

- **Options for a contracted TB bladder:** augmentation cystoplasty (first described in the 19th century for this condition) and bladder substitution.
- **Augmentation indication:** intolerable frequency, nocturia, urgency, pain and haematuria, typically with capacity < 100 mL.

BOWEL SEGMENT BY DEGREE OF BLADDER CONTRACTION

Bladder state	Procedure / segment
Half the bladder diseased	Augmentation with ileum
Severely contracted	Augmentation with ileocaecum or sigmoid (stomach and caecum also used)
Thimble bladder < 20 mL	Orthotopic bladder substitution

- **Usual rules for bowel in the urinary tract apply:** assess renal function, reconfigure bowel into a low-pressure reservoir, educate the patient, ensure long-term follow-up.
- **Complications:** mucus, electrolyte disturbance, secondary bacterial infection.
- **Laparoscopic and robot-assisted augmentation:** increasing use, good results.

Urethral Procedures

- **Bladder neck contracture** → transurethral incision.
- **Urethral stricture** → endoscopic treatment, often repeated.
- **Tuberculous urethral fistula** → chemotherapy + suprapubic drainage, then delayed reconstruction.
- **A tuberculous seminal vesicle cavity** has been drained into the bladder by cold-knife incision (case report).

Genital Surgery

- **Extirpative genital surgery** is reserved for failed medical therapy.
- **Epididymis involved, testis spared** → epididymectomy alone, not orchidectomy, preserving testicular blood supply: ligate the vas → begin dissection at the globus minor → excise.
- **Testis involved** → scrotal orchidectomy.
- **The vas is usually diseased distal to the external ring**, so ligation at the ring is possible and sufficient.

Monitoring for Tuberculosis Relapse

- Relapse after pulmonary TB: 2–6%, mostly within the first year; needs a second, longer or different regimen.
- Relapse after UG TB is higher: 6.3–22%, even after 12 months of treatment.
 - Reason: an extensively diseased kidney holds innumerable foci that are hard to sterilise; viable bacilli have been found after 9 months of treatment.
- In any recurrence, make a special effort to isolate the organism for susceptibility testing.
- Follow-up: pulmonary TB 2 years; UG TB, some recommend 10 years, as mean time to relapse is about 5 years (5.3).

Concept — Surgery in UG TB follows chemotherapy. It is timed at 4–6 weeks, aims first to preserve the kidney, and is mostly reconstructive. It exists because drugs sterilise but cannot reverse fibrosis. The poorly sterilised kidney is also why UG TB relapses more often and needs longer follow-up than pulmonary TB.

Management of Urogenital Tuberculosis in Special Situations

- In every special situation, choose the regimen carefully (side effects, interactions, toxicity) and seek infectious diseases or TB specialist advice.

Multidrug-Resistant, Extensively Drug-Resistant and Pre-Extensively Drug-Resistant Tuberculosis

DRUG-RESISTANCE DEFINITIONS (UNITED STATES)

Category	Resistant to
MDR-TB	INH + rifampicin (the two most important first-line drugs)
Pre-XDR-TB (definition since 2022)	INH + rifampicin + a fluoroquinolone OR an injectable aminoglycoside
XDR-TB	INH + rifampicin + a fluoroquinolone + either an injectable second-line aminoglycoside (amikacin, kanamycin, capreomycin) OR bedaquiline or linezolid

OUTCOME BY RESISTANCE CATEGORY

	Drug-susceptible	MDR-TB	XDR-TB
Cure rate	94–97%	50–60%	30–50%
Conventional regimen	6 months	Usually > 18 months, sometimes with IM drugs	5–6 drugs for ≥ 2 years

- Frequency:
 - Worldwide: MDR-TB in about 3.6% of new and 20% of previously treated cases; one-fifth of MDR-TB has additional resistance.
 - United States: < 1.5% at first diagnosis, stable over 20 years.
- BPaL (bedaquiline + pretomanid + linezolid), approved since 2019: all-oral MDR-TB regimen (duration and extrapulmonary status in table below), used successfully in XDR-TB without bedaquiline or linezolid resistance; role in UG TB not established.

Pregnancy and Lactation

- Advise avoiding pregnancy during treatment for active TB.
- TB diagnosed in pregnancy → treat promptly: fetal risk from TB outweighs drug risk.

PYRAZINAMIDE IN PREGNANCY — TWO POSITIONS

United States	WHO
Controversial: fetal effects not established, perceived higher hepatotoxicity risk → usually omitted , so treatment lasts 9 months	Recommended as part of the standard first-line regimen

- **Breastfeeding is allowed once the mother is non-infectious; breast-milk drug levels are too low to be toxic.**

Human Immunodeficiency Virus Infection

- **HIV raises active-TB risk 15- to 20-fold; each infection accelerates the other. Test every TB patient for HIV.**
- **TB causes > 30% of deaths among PLHIV worldwide, reminiscent of 18th- and 19th-century Europe.**
- **Extrapulmonary and UG TB are commoner in HIV; UG TB was common at autopsy in AIDS in a small Indian study.**

UG TB — HIV-POSITIVE VS HIV-NEGATIVE

	PLHIV	HIV-negative
Extent	More disseminated; more lymphadenopathy; bilateral renal disease	More localised
Caseation, necrosis, fibrosis	Less (fibrosis needs a competent immune response)	Typical
Collecting-system stenosis	Much less frequent	Most patients
Bladder contracture	Much less frequent	Common
Mortality	High despite fewer obstructive lesions	Lower

- **Treatment:**
 - **Do not delay.** Guidelines **match those without HIV: the 6-month short course is effective; 9 months no longer routine.**
 - **Duration depends on site, severity, tolerance, response and whether antiretroviral therapy is effective.**
 - **Interactions: rifamycins lower antiretroviral levels (rifampicin more than rifabutin), so antiretroviral doses may need increasing.**

Concept — In HIV, UG TB causes less fibrosis because fibrosis is an immune product. The disease is therefore more disseminated and more lethal but less obstructive. The inverse relationship between immunity and scarring runs through the whole TB half of the chapter.

Renal Transplant Recipients

- **Allograft TB is rare: usually within 6 months of transplantation, up to 7 years.** Early onset reflects immunosuppression or TB already in the donor kidney.
- **Diagnosis is difficult:**
 - Early disease, so **imaging usually normal.**
 - Immunosuppression **blunts the usual pathology.**
 - **Fever is the usual presentation; urinary symptoms in only a minority (20%).**
- **Diagnostic yields:**
 - Chest radiograph: **abnormal but non-specific in about half.**
 - Sputum culture: **positive in about half**, whatever the chest radiograph shows.
 - **Urine AFB culture: positive in all patients in one study — the key test.**

- Many are diagnosed only on histology after graft nephrectomy.
- **Treatment:** rifamycin-immunosuppressant interactions require frequent drug-level monitoring and dose adjustment.
- **Complications:** graft rejection, disseminated TB, death in up to about one-third (36%).

SPECIAL SITUATIONS — WHAT CHANGES

Situation	Regimen change	Duration
MDR-TB	Second-line drugs; BPaL (not yet approved for extrapulmonary TB)	> 18 months conventional; 6–9 months with BPaL
XDR-TB	5–6 drugs	≥ 2 years
Pregnancy	INH + rifampicin + ethambutol + pyridoxine; pyrazinamide omitted in the United States	9 months
HIV on effective antiretrovirals	Standard; daily dosing; watch rifamycin interactions (rifabutin preferred)	6 months
HIV not on effective antiretrovirals	As above	9 months
Renal transplant	Rifamycin dose adjustment with drug levels, or rifamycin-free regimen	≥ 18 months if rifamycin-free

KEY POINTS: TUBERCULOSIS

- **Untreated UG TB causes irreversible damage (renal failure, infertility)**, so TB must be in the differential diagnosis of UG disorders.
- **Presentation is non-specific and indolent**, and constitutional symptoms are infrequent. **A high index of suspicion is essential.**
- **Gold standard: positive AFB culture from urine or tissue biopsy.** A diagnosis can often be made by other means, such as **NAAT** or **granulomas on histology**.
- **Optimal timing of surgery: 4–6 weeks after starting medical therapy**, so that inflammation settles, the bacillary load falls and the lesions stabilise.
- **Monitor for relapse after treatment. It occurs in up to 22%.**

PARASITIC INFECTIONS OF THE UROGENITAL TRACT

- Parasites are rare in non-endemic practice, but the urologist must recognise them to diagnose and treat correctly.
- **Relevant parasites:** UG schistosomiasis, filariasis, amoebiasis, enterobiasis and echinococcosis.

Schistosomiasis

- More than 200 million people are infected with *Schistosoma* species.

THE THREE MEDICALLY IMPORTANT SCHISTOSOMES

Species	Distribution	Principal target	Snail host
<i>S. haematobium</i>	Africa, Arabian Peninsula	Urogenital tract	<i>Bulinus</i>
<i>S. mansoni</i>	Africa, Arabian Peninsula, South America	Liver, GI tract	<i>Biomphalaria</i>

Species	Distribution	Principal target	Snail host
<i>S. japonicum</i>	China, South-East Asia	Liver, GI tract	<i>Oncomelania</i>

- **UG schistosomiasis is the *S. haematobium* disease:** a complex life cycle, many manifestations and close ties to the environment.
- **It has co-evolved with humans and non-human primates for millennia**, so ancient civilisations recognised it.

History

- **Schistosome antigens in Egyptian mummies (c. 3500 BCE); *S. haematobium* eggs** in later mummies.
- **The Egyptians called it "A-a-a disease";** the hieroglyph shows a **penis dripping blood-stained urine**.
- **Theodor Bilharz (1852, Cairo autopsies)** found worms in the **mesenteric veins** and linked them to **eggs in urine and stool** — hence "bilharzia".

Biology and Life Cycle

- **Human cycle:**
 - **Cercariae penetrate skin (even intact)** in infested fresh water. They live about **1 day** and lose infectivity **within hours** of leaving the snail.
 - They **shed their tails over 90–120 minutes** → **schistosomulae**.
 - Schistosomulae migrate via blood or lymph **to the lungs** (several weeks) → venous circulation → **liver sinusoids**, where they **begin feeding on blood**.
 - **Adults migrate to the vesical and pelvic venous plexus** and live on average **3–5 years**. Cercaria to adult takes **80–110 days**.
 - **Pairing:** the male holds the female in the **gynaecophoric canal**, helps her pump blood and **chemically stimulates egg-laying**.
- **Eggs:**
 - **Fecundity:** hundreds to thousands of eggs per female per day.
 - ***S. haematobium* eggs: oval, about 140 µm, terminal spine** (Fig. 31.9).
 - Eggs must **cross the endothelium and bladder wall** to reach urine; some cross the gut wall into faeces.
 - **Fewer than half are excreted; eggs retained in tissue drive the pathology.**
 - Eggs survive about **20 days if kept wet**.
- **Environmental cycle:**
 - **Fresh water and light** → eggs hatch → **ciliated miracidia** (live about **6 hours**).
 - Miracidia penetrate ***Bulinus*** snails, which prefer **slow-flowing water** and tolerate **low oxygen**.
 - **Primary sporocyst** → **20–40 daughter sporocysts** → **200–400 cercariae each**, released into water.
 - **The prepatent period is temperature-dependent; no cercariae are shed below 15 °C or above 35 °C.**

FIG. 31.8 Life cycle of a schistosome. Source: King CH, *Schistosomiasis*, in *Tropical Infectious Diseases*, 2006. Points the figure adds to the text:

- Cercariae are forked-tailed (about 500 µm) and cause "**swimmer's itch**" on penetration.
- Adults are **1–3 cm long** and pair for life.
- **Eggs measure 70–170 µm** across species; miracidia hatch within 1–3 hours.

- **Each species has its own venous destination and snail:**
 - *S. haematobium* → **vesical venous plexus**: haematuria, bacterial UTI, scarring, calcification, **SCC**.
 - *S. mansoni* and *S. japonicum* → **mesenteric venules of bowel and rectum**: **Katayama fever** (especially *S. japonicum*), **presinusoidal granulomas**, **Symmers pipestem periportal fibrosis**, portal hypertension.
 - All species occasionally cause **embolic egg granulomas in the brain or spinal cord**.

FIG. 31.9 Micrographs of *S. haematobium* eggs showing the **characteristic terminal spine**. Source: Ray et al., *PLoS Negl Trop Dis*, 2012.

Epidemiology

- **Distribution follows the snail: most of sub-Saharan Africa, parts of North Africa and the Middle East.**
- **Europe — Corsica, 2013: the first autochthonous transmission in Europe for over a century; further cases in 2015 suggest a new endemic focus.** Some were *S. bovis* or *S. haematobium*–*S. bovis* hybrids.
- **Burden:**
 - **Mostly the UG form**; newer estimates reach **440 million** infected.
 - **Up to 150,000 deaths a year from *S. haematobium* obstructive renal failure alone.**
 - In sub-Saharan Africa, tens of millions have haematuria and dysuria at any time; **millions have major bladder pathology and hydronephrosis.**
- **Individual risk depends on the type and duration of freshwater contact:**
 - **Rural women and children:** domestic chores (laundry, dishwashing).
 - **Children:** play and swimming.
 - **Adults:** fishing, car washing, agriculture with **water-intensive crops (rice, sugar cane).**
 - **Prevalence is highly focal**, because transmission depends on water.
- **Dams and irrigation increase transmission** by creating **year-round slow-flowing snail habitats** and drawing **denser farming populations**.

DAMS AND SCHISTOSOMIASIS

Dam	Decade	Effect
Aswan Low Dam, Egypt	1930s	Flood → perennial irrigation; prevalence rose from < 11% to > 75% in some regions
Aswan High Dam, Egypt	1960s	Similar rise
Dama Dam, West Africa	1980s	Similar rise

- **Highest prevalence: poor sanitation and no clean running water.**
- **Children aged 5–15 carry the heaviest worm burden**; whether from exposure or susceptibility is unclear.
- **Schistosomiasis and poverty reinforce each other:** the **snail hosts live in tropical and subtropical developing countries**; **poor sanitation** puts eggs into surface water; **no safe water supply** prolongs water contact; and **chronic infection impairs growth, learning and productivity**, perpetuating poverty.

Concept — Schistosomiasis is an environmental disease. Transmission needs human excreta in fresh water, a particular snail and prolonged contact with water. Every feature of its epidemiology (focality, children, dams, poverty) is one of those three links, and so is every method of control.

Pathogenesis and Pathology

- **Cercarial penetration** is aided by secreted **proteases**, which trigger cellular and humoral responses; but the parasite **modulates host immunity from first contact**, so brief penetration causes **only local skin inflammation**. **Repeated exposure → hypersensitivity → maculopapular rash**.
- **Immune evasion by the tegument**: the maturing schistosomulum forms a **double lipid bilayer surface (the tegument)** that lets it survive for years, via **host antigen mimicry, continuous membrane turnover, immunomodulatory proteins and proteases, evasive biophysical surface properties** and **modulation of surface-antigen expression**.
- **Most knowledge comes from autopsy studies and animal models**.

WORM VS EGG — THE GOVERNING CONTRAST

Adult worm	Egg
Immunologically near-silent (tegumental evasion)	The main immunopathological trigger
Lives in the venous plexus	Deposited in the walls of the bladder and pelvic organs
Causes little disease directly	Granuloma → fibrosis → obstruction of blood or urine flow

- **The egg granuloma**: a **mixed infiltrate of eosinophils, plasma cells and lymphocytes**. **Continuous egg-laying means granulomas of every stage coexist**; clustered eggs form **composite, coalescing granulomas**, grossly **bulky, hyperaemic, polypoid masses** projecting into the bladder (Fig. 31.10).
- **Severity modifiers**: **host genetics, in utero sensitisation, co-infection**.
- ***S. haematobium* eggs rapidly induce type 2 inflammatory genes and suppress urothelial barrier genes**, suggesting that **host and parasite share the aim of expelling eggs** across a temporarily weakened urothelium.

FIG. 31.10 IVU showing **scalloping of the bladder and right lower ureter** by schistosomal polypoid lesions.

- **Ectopic egg deposition**:
 - **Portal tract**: heavy egg loads swept to the liver block **presinusoidal capillaries** → granulomas → **portal hypertension**. Embolised eggs give **granulomatous, fibrotic portal tracts** and **dilated portosystemic collaterals** → **pipestem (Symmers) fibrosis** → **hepatosplenomegaly**. Susceptibility depends on the individual immune response.
 - **Lung**: worm pairs reach the pulmonary vessels **only in very severe infection**, after pelvic disease is established → **pulmonary fibrosis, pulmonary hypertension, cor pulmonale**.
- **Naturally acquired immunity exists**: some stay **egg-negative for ≥ 5 years** despite continued exposure and no treatment.

RESISTANT VS CHRONICALLY INFECTED

Resistant to reinfection	Chronically infected
Mixed T-helper 1 (Th1) + Th2 response	Th2 response only
Protective IgE active	IgG4 blocks protective IgE (antibodies to worm and egg antigens)

- **IgE to worm antigens rises with age**, suggesting immune resistance — but **water contact also falls with age**. Exposure-controlled analyses still suggest **age contributes independently**.
- **Calcification and obstructive uropathy — the causal chain**:
 - **Unexpelled eggs calcify** (bladder, ureters).
 - → **reduced urinary tract compliance** → **higher upper-tract pressure** → **stasis, hydroureter, hydronephrosis**.

- **Calcification extent on imaging roughly reflects the tissue burden of calcified eggs.**

LEVEL OF SCHISTOSOMAL URETERIC OBSTRUCTION

Level	Frequency
Ureteric meatus	~1%
Intramural (interstitial) ureter	10–30%
Juxtavesical ureter	20–60%
Lower third of ureter	15–50%
Contiguous combination	30–60%

- **The pattern:** obstruction is **overwhelmingly distal** (juxtavesical to lower third); the meatus itself is rarely the site.

THREE PATTERNS OF SCHISTOSOMAL HYDROURETER

	Segmental (cylindrical or fusiform)	Tonic	Atonic
Share of obstructive uropathy	About one-quarter (80% in the lower ureter)	Up to 30%	The remainder
Wall	Dilatation above concentric muscle replaced by fibrosis and sandy patches	Thick-walled, trabeculated; marked muscular hypertrophy; tortuous	Thin-walled; atrophic, fibrotic muscle; very tortuous
Peristalsis	—	Impaired	Absent
Extent	Segmental	Whole ureter above the lesion (functional stenosis)	Markedly dilated
Hydronephrosis	Rarely significant	Significant, but reversible if obstruction is relieved	—

- **The logic:** tonic = **hypertrophied muscle fighting an obstruction** (reversible); atonic = **muscle that has failed** (fibrotic, aperistaltic). The trap is assuming every dilated ureter is irreversibly damaged.
- **Hydroureter precedes hydronephrosis.** Untreated hydronephrosis progresses **pelvic dilatation → medullary atrophy → medullary effacement → cortical atrophy**, which is why **tubular function** (concentrating ability) is lost before glomerular function.
- **Bacterial UTI is commoner in chronic infection**, possibly from **bacteria adhering to the worm tegument, stasis and immunomodulation**.
- **Bladder cancer:** *S. haematobium* has the **strongest cancer link of any helminth**, particularly to **SCC**, on epidemiological (especially **Egypt**) and experimental evidence. **Risk rises with duration and severity of infection**.
- **Female genital schistosomiasis (FGS):** egg sequestration causes **fibrotic nodules ("sandy patches")** in the **uterus, cervix and lower genital tract** (Fig. 31.11). The mechanism is poorly understood beyond **increased vascularity of the genital mucosa** around eggs.
- **Male genital schistosomiasis (MGS):**
 - **Cause:** ectopic eggs in the **testis, epididymis, spermatic cord, scrotum and penis**.
 - **Poorly characterised**, partly because **transrectal US is needed** to assess the internal sex organs.
 - **Pathology:** **neoplastic growth around lesions** with fibrosis and calcification.

- **Presentation:** genital disfigurement, haemospermia, haematuria, infertility, genital pain; a minority have sexual dysfunction or urethral discharge mistaken for a sexually transmitted infection.
- **Infertility is thought reversible** (epididymitis, funiculitis); azoospermia with eosinophilic interstitial orchitis can occur.
- **Diagnosis:** eggs in the ejaculate on microscopy. But symptoms predict disease poorly, semen egg counts vary between samples and correlate poorly with urine counts, and the effect of abstinence is unknown.
- **Hydrocele and testicular calcification** are reported; causal link uncertain.
- **Raised PSA** suggests frequent prostatic involvement. **Serum testosterone is lower** with *S. haematobium* or *S. mansoni* infection (mechanism unknown).
- **MGS, like FGS, may increase HIV susceptibility.**
- **Treatment:** praziquantel.
- **Bladder sequelae:** urothelial hyperplasia → squamous metaplasia → dysplasia → urothelial carcinoma or SCC. Bladder cancer is the final pathological sequel.

SCHISTOSOMAL VS CONVENTIONAL BLADDER CANCER

	Schistosomal bladder cancer	Conventional (non-endemic) bladder cancer
Age at onset	Early: 40–50 years	Older
Histology	SCC 60–90%; adenocarcinoma 5–15%	Predominantly urothelial (transitional cell)
Differentiation	> 40% of SCCs well differentiated, verrucous, good prognosis	—
Site	Posterior wall about half; lateral wall about 30%	—
Growth	Exophytic about two-thirds; rest ulcerative, endophytic	—

- **Mass drug administration (MDA) in Egypt reduced bladder neoplasms and shifted SCC to transitional cell carcinoma** — evidence of cause.
- **Smoking**, common in endemic regions, may act **synergistically**; yet some autopsy series there show similar cancer rates without schistosomiasis.
- **Carcinogenesis:**
 - The International Agency for Research on Cancer (IARC, WHO) classifies *S. haematobium* as Group 1 ("carcinogenic to humans").
 - **Vascular endothelial growth factor (VEGF)** rises in the bladder soon after egg exposure and may promote tumour vasculogenesis and progression.
 - **Proposed pathway:** papillomas merge with basal transitional epithelium → **benign fibroepithelial papillary growths** → repeated inflammation and fibrosis → **clonal, precancerous lesions including squamous metaplasia.**
 - Egg exposure alters **carcinogenesis-related signalling pathways** (mouse bladder).
 - **Co-infecting bacteria convert nitrates to nitrosamines**, which **alkylate DNA and protein**, causing mutations in genes such as **p53**.

FIG. 31.11 Macroscopic urinary schistosomiasis:

- **(A) Early active stage:** bladder opened by anterior Y incision; posterior and apical walls carry many erythematous, granular, sessile and pedunculated polyps.

- **(B) Chronic active and inactive foci:** coronal section of the apex. Lamina propria replaced by a **yellow-tan, finely granular sandy patch** (chronic inactive); small sandy patches scattered through **fibrotic, atrophic detrusor** and even perivesical fat. Superficial lamina propria still holds **viable eggs with granulomas** (chronic active).
- **(C) Chronic inactive stage with the bilharzial bladder cancer syndrome:** lamina propria replaced by a **concentric sandy patch**, most prominent at the edge of an **exophytic, moderately differentiated SCC**; no recent egg-laying.
- **(D) Severe chronic inactive disease:** whole lamina propria replaced by sandy patch, with foci of **epidermisation**. One ureteric orifice is a **dilated "golf-hole" (schistosomal uropathy)**; the other **markedly stenotic**.
- **(E) Rectosigmoid polyposis:** erythematous polyps with **active egg-laying**; some have necrotic, haemorrhagic tips.
- **(F) Partial cystectomy specimen (chronic inactive stage):** a **stellate chronic schistosomal ulcer**. Legend truncated here; it ends in a reference to metaplasia.

Concept — *In schistosomiasis the worm is tolerated and the egg is not. Eggs held in the wall give granuloma, then fibrosis and calcification, then obstruction, and eventually metaplasia and cancer. Every urological sequel is the egg's damage accumulating over time.*

Clinical Manifestations

ACUTE VS CHRONIC SCHISTOSOMIASIS

	Acute	Chronic
Cause	Response to cercarial penetration and worm migration and maturation	Immune response to prolonged egg-laying over years
Who	Often travellers , in primary infection	Long-term endemic residents: repeated reinfection, high worm burden
Outcome	Self-limiting	Organ damage

- **Schistosomes cannot multiply in the human host**, so after a single exposure **adults die of old age within 3–5 years** even untreated, limiting pathology in visitors.

Acute Schistosomiasis

- **Cercarial dermatitis:** **itchy maculopapular rash within 1–2 days** of penetration; it **often resolves before the traveller returns home**, hindering diagnosis.
- **Katayama fever (acute schistosomiasis):**
 - **2–8 weeks after primary infection** in some patients (many are asymptomatic).
 - **Named after the Katayama Valley, Japan.**
 - Commonest with heavy *S. japonicum*, less common with *S. mansoni*, **rare with *S. haematobium*.**
 - **Features:** **fever, dry cough, fatigue, headache, diarrhoea, eosinophilia, neck pain, urticaria.**
 - **Rare in endemic residents.**
 - **Non-specific:** often missed or mistaken for **malaria or enteric fever.**
- **Because acute infection can be silent, anyone exposed to potentially infested water should be considered for testing and treatment.**

Chronic Schistosomiasis

- **Egg-laying into the bladder begins 8–12 weeks after infection**, presenting with **painless, recurrent haematuria**, dysuria or frequency.
 - In highly endemic cultures **male haematuria is seen as a sign of puberty**; it can cause **anaemia**. **Proteinuria** is common.
 - **Haematuria is consistent and specific enough to serve as a primary diagnostic test in endemic areas**, but in **returning travellers** is often not recognised as schistosomal.
- **Ureteric disease**: ureteric eggs → granulomas, polyps, ulcers → **impaired pelvic and ureteric peristalsis** → **hydronephrosis, hydroureter, obstruction and vesicoureteric reflux**.
 - **Short-term infection**: renal function can recover with anthelmintics.
 - **Late or severe disease**: ureteric repair or urinary diversion may be needed.
- **FGS affects one-third to three-quarters of women infected with *S. haematobium***; eggs lodge in the fallopian tubes, cervix, vagina, vulva, ovaries and uterus.
 - **Friable sandy patches bleed on contact** (examination, intercourse).
 - **Dyspareunia, pelvic pain, bleeding, discharge, frequency and infertility** mimic UTI and sexually transmitted infection → **misdiagnosed and untreated**.
- **Male genital involvement**:
 - Heavy egg loads in the **ejaculatory ducts and seminal vesicles**; **blood or eggs can appear in the ejaculate before eggs appear in urine**.
 - Presents as a **testicular mass or scrotal pain**.
 - **Egg burden is higher in the epididymis, ovaries and tubes than in the testis, uterus and vagina**.

STAGES OF CHRONIC URINARY SCHISTOSOMIASIS

Stage	Egg activity	Clinical picture
Active	Eggs being laid; haematuria	Polypoid lesions, haematuria
Late chronic active	Tissue egg burden peaks	"Schistosomal contracted bladder" : chronic suprapubic and pelvic pain, urgency, frequency, incontinence. Trigone normal or mildly hyperaemic; rest of detrusor thickened, indurated. Functional capacity as low as 50 mL
Quiescent	Egg-laying and excretion fall; symptoms fade (> 30% of light infections become asymptomatic)	Silent obstructive uropathy continues : fibrosis replaces polyps → insidious severe hydroureteronephrosis
Chronic inactive	No viable eggs in urine or tissue	Sequelae of reaction to calcified dead eggs . 40–60% with schistosomal obstructive uropathy first reach a urologist at this end stage . Poorly or non-functioning kidneys in asymptomatic people

- **The trap: a normal urine egg count does not exclude severe disease**. The most damaged kidneys belong to inactive-stage patients, who have no eggs and often no symptoms.
- **Bacterial superinfection**:
 - **About half** with obstructive uropathy develop superimposed UTI from the **usual uropathogens**.
 - ***Salmonella* association**: chronic or recurrent *Salmonella* UTI with **intermittent bacteraemia**; bacteriuria may be **spill-over from the bloodstream**. *Salmonella* shelters in the **apical folds of the worm tegument**, protected from host defences and antibiotics.
 - **Treat both infections; antibiotics alone do not clear it**.

- **Bladder ulcers:**
 - **Acute ulcers are rare**, forming in the active stage when **necrotic polyps slough**.
 - **Chronic ulcers are commoner**, a **late sequel of heavy infection**, with **constant burning**, intense suprapubic and pelvic pain, usually gross haematuria and pyuria.
- **Eosinophilia: very common in acute, low-grade in chronic infection**; neither sensitive nor specific, but a clue to parasitic infection.
- **Exceptions to the usual sequence:** ectopic **pulmonary schistosomiasis**, **neuroschistosomiasis** and **FGS**.

BLADDER — TB VS SCHISTOSOMIASIS

	TB	Schistosomiasis
Pathogenesis	Descending infection from the kidney	Egg deposition in the bladder wall
Trigone	Spared (dome most affected)	Normal or mildly hyperaemic (rest of detrusor indurated)
Contracted bladder	Thimble bladder < 20 mL	Functional capacity as low as 50 mL
Ureteric orifice	Golf-hole	Golf-hole (dilated) or stenotic
Ureteric calcification	Intraluminal, non-dilated ureter (forms a cast)	Intramural, dilated ureter
Malignancy	SCC of the pelvis after metaplasia (rare)	SCC 60–90% of schistosomal bladder cancers

Concept — Chronic schistosomiasis dissociates symptoms from damage. Egg-laying and symptoms decline over the years while fibrosis goes on obstructing the tract. The worst kidneys are therefore found in quiet, egg-negative patients in the inactive stage.

Diagnosis

- **Gold standard for active infection: *S. haematobium* eggs in urine or stool** (see table for timing).
- **Urine sampling: egg excretion peaks at noon**, so collect between 09:00 and 15:00; **concentrate the urine** to detect light infection.
- **No eggs but high suspicion, with serology showing exposure → tissue biopsy:**
 - **Rectal snip first**, before bladder biopsy: **eggs are common in rectal mucosa** and the **risks of bladder biopsy (infection, perforation) are avoided**.
 - **A squash preparation between glass slides beats histology: more sensitive and shows egg viability.**
- **Suspected FGS: biopsy vulval, vaginal or cervical lesions** for eggs.
- **Indirect markers (visual or dipstick haematuria, urine turbidity)** are used mainly in **control and elimination campaigns**, combined with other tests.

DIAGNOSTIC TESTS FOR UG SCHISTOSOMIASIS

Test	Indicates	Strength	Limitation
Urine or stool egg microscopy	Active infection (gold standard)	Direct proof	Negative in the first 8–12 weeks ; insensitive in light infection
Rectal snip / bladder biopsy (squash preparation)	Active or past infection	Egg viability	Invasive
Serology: FAST-ELISA + Western blot (CDC)	Exposure	> 90% sensitive and specific ; positive from 4–6 weeks	Cannot distinguish acute, chronic or cured ; titres persist after cure

Test	Indicates	Strength	Limitation
Circulating anodic and cathodic antigens (CAA, CCA), serum or urine	Active infection only (released only by live adult worms)	Quantitative (intensity); point-of-care urine CCA strip	CCA strip has failed to detect <i>S. haematobium</i> (good for <i>S. mansoni</i>); expensive
PCR (urine, stool, vaginal lavage for FGS)	Infection	Most sensitive method	Needs trained staff, solvents, kits; impractical in the field; useful for surveillance in low-transmission settings
Haematuria dipstick	Indirect	Cheap, used in campaigns	Low sensitivity and specificity

- What the table is for: sorting tests by what a positive means. Only eggs and CAA/CCA prove active infection; serology proves exposure only. The trap is using serology to judge cure.

- Imaging:

IMAGING IN UG SCHISTOSOMIASIS

Modality	Findings and use
US	Bladder and ureteric wall thickening, polyps, hydroureter, hydronephrosis, calcification, even bladder carcinoma
Plain film	Urinary tract calcification. Calcified bladder like a "fetal head in the pelvis" is characteristic of chronic disease. Prostate, seminal vesicles, posterior urethra, distal ureters and occasionally colon can also calcify
IVU	Earliest sign: striations of ureters and renal pelvis. Also hydronephrosis, hydroureter, non-function, stenosis, polypoid filling defects. Delayed films in severe obstruction; post-void views show bladder neck obstruction. IVU + fluoroscopy distinguishes tonic from atonic ureters and shows non-stenotic immobile ureters
CT	Obstruction plus colonic and urinary tract calcification — an advantage over IVU
MRI	No proven superiority; not routine
Voiding cystourethrography	Vesicoureteric reflux in about 25% of infected ureters
Cystourethroscopy ± retrograde pyelography	Mucosal lesions (Fig. 31.13); ureteric anatomy and drainage

FIG. 31.12 Bladder calcification. (A) Plain abdominal film: rim of calcification around the bladder. (B) Ultrasound: bright line around the bladder with dark acoustic shadow behind. Source: Abdel-Wahab et al., *J Urol*, 1992.

FIG. 31.13 Cystourethroscopy in a child from Somalia with intermittent haematuria and dysuria: papillary mucosal and submucosal bladder lesions.

- Globally, most cases go undetected and untreated because diagnosis relies on direct egg detection rather than more sensitive methods; better accessible diagnostics are needed for control.

Treatment

Medical Management

- Praziquantel (PZQ) is the only WHO-recommended drug. Single-drug dependence risks resistance, but PZQ is effective, widely available and of low toxicity, so there is little incentive for alternatives.
- Dose for *S. haematobium*: 20 mg/kg twice on the same day, 6–8 hours apart, or 40 mg/kg once.
- Katayama fever: add corticosteroids.

- **Efficacy (egg reduction and cure rate): 60–90%.** Even without cure, **worm burden falls substantially**, reducing further sequelae.
- **Follow-up: urine and stool egg counts and serial UG ultrasound; repeat the course** if infection persists.
- **Tolerability: mild side effects** (abdominal pain, nausea, headache, dizziness) **within 3–4 hours**, settling spontaneously. **Large, bitter tablets** are difficult for children; a paediatric formulation is in development.
- **Pregnancy: FDA Category B** (animal safety data only), so many programmes **exclude pregnant and lactating women** — yet **adverse effects are rarely reported despite millions of pregnant women treated**.
- **Weak activity against schistosomulae** means lower cure rates where **transmission and reinfection** are high, and **PZQ cannot abort infection soon after exposure**. **Repeat doses several weeks apart** catch juveniles missed earlier that have since matured.
- **Resistance: little in endemic areas in large studies**; failures reported in **returning travellers and military personnel**. **Protection: alternative drugs and environmental control** (snail control, sanitation).

PRAZIQUANTEL VS ARTEMISININ DERIVATIVES

	Praziquantel	Artemether / artesunate
Stage targeted	Adult worms (weak against schistosomulae)	Schistosomulae
Role	Treatment of established infection	Chemoprophylaxis (90–97% effective); poor treatment of established infection
Combination	PZQ + artemisinin → lower infection rates than PZQ alone (useful for MDA in high-transmission areas)	
Caution	—	Avoid wide use in malaria co-endemic areas (risk of artemisinin-resistant malaria)

- **What the pair tests:** the drugs are **complementary by life-cycle stage**; together they cover juvenile and adult worms. The limit is **malaria**, not schistosomiasis.

Surgical Management

- **Medical therapy first:** PZQ is effective and easy to give, and **early disease can reverse**.
- **Surgery is for complications unresponsive to adequate medical treatment** after reasonable follow-up, or those needing **immediate intervention** — e.g. **severe bladder haemorrhage**, a common cause of urgent surgery.
- **Bladder outlet: prostatitis and prostatic enlargement are uncommon** and **autopsies show no anatomical outlet obstruction**; but **cystoscopic and urodynamic findings and raised post-void residuals show functional outlet obstruction**, occasionally needing surgery in severe inactive disease.
- **Scrotum: induration, pain and enlargement from schistosomal epididymitis** can prompt **surgery for suspected testicular tumour**.
- **Irreversibly contracted bladder: vesical denervation, urinary diversion, ileocystoplasty or hydrodistension**. Always combine with chemotherapy.
- **Chronic deep bladder ulcer → partial cystectomy**; fulguration rarely relieves symptoms or heals the ulcer.
- **Urothelial hyperplasia** accompanies **severe disease**; **dysplasia and metaplasia** accompany **schistosomal bladder cancer**, which is managed surgically (see bladder cancer chapters).

- **Obstructive uropathy is the commonest sequel**; hydroureter and hydronephrosis track **infection intensity**.

URETERIC OBSTRUCTION — ACTIVE VS LATE DISEASE

Active infection	Late chronic / inactive disease
Concentric or hemi-concentric polypoid lesions "girdle" the intramural and adjacent extravascular ureter	Anatomical stenosis ($\pm$ calculi) in up to 80% of obstructed ureters
Responds to medical treatment alone. Renal function recovers within 1–2 months of PZQ. Chemotherapy also prevents obstruction despite reinfection	Less responsive to chemotherapy, so persistent stenosis needs surgery

SURGERY FOR RESIDUAL SCHISTOSOMAL URETERIC STENOSIS

Lesion	Procedure	Note
Anatomical stenosis	Balloon dilatation	Effective, but mechanical dilatation often restenoses
Meatus, intramural ureter, vesicoureteric junction, distal ureter	Reimplantation with a functional valve: Politano–Leadbetter variants	Highly effective in some; restenosis can occur
Long or multifocal lesions (too little ureter for reimplantation or ureteroureterostomy)	Boari flap; ileal conduit; suprapubic intravesical ureterostomy (peritoneal dialysis catheter bypasses the segment into the bladder); ileal ureter	Keep the ileal segment isoperistaltic
Isolated meatal stenosis	Meatoplasty	—
Unreconstructable ureter	Long-term nephrostomy	—

Prognosis

- Most have mild infection and a good prognosis. Morbidity and mortality depend on infection intensity and immune-response gene polymorphisms.

PROGNOSIS BY PREVALENCE SETTING

Low-prevalence region (e.g. Nigeria)	Egypt at 50% prevalence
Essentially no related mortality; little obstructive uropathy	Schistosomiasis contributed to death in 10% of infected
	Severe disease: mortality approached 50% within 2–5 years

TWO DEATH PROFILES

Death from obstructive uropathy	Death from pyelonephritis or urothelial cancer
Patients in their 20s with heavy egg burdens	Patients over 40
Bilateral end-stage hydronephrosis, or unilateral with contralateral non-schistosomal end-stage renal disease	Reflects time- and intensity-related pathology

- PZQ has transformed prognosis: in children, obstructive polyps usually resolve completely within 2–6 weeks.
- Chronic obstruction from sandy patches and fibrosis: prognosis less certain; some tolerate advanced uropathy with little renal decline.

- **Obstructive uropathy, stones, outlet obstruction and bacterial cystitis all predispose to pyelonephritis. Bacterial superinfection can be life-threatening — treat promptly and aggressively.**
- **Bladder cancer:** prognosis depends on tumour aggressiveness.

Prevention and Control

- **Travellers: avoid fresh water** (streams, rivers, ponds, lakes); **fast-flowing water can still transmit.**
- **Cercariae are killed by heating water to > 125 °F (about 52 °C) for 5 minutes, chlorination, or standing water > 2 days, free of snails.**
- **PZQ was introduced in the late 1970s, with mass distribution from the 1980s and 1990s; treatment is simple and cheap, but control remains hard.**
- **WHO strategy: periodic, usually annual, targeted MDA with PZQ, aimed at reducing morbidity. Without safe water, cycles of infection, treatment and reinfection demand more frequent dosing.**
- **Cost: under US\$0.50 per person in campaigns, yet at scale this can exceed national budgets; donated PZQ from companies and foundations helps.**

BREAKING TRANSMISSION — THREE CONTROL LEVERS

Lever	Mechanism	Limitation
Sanitation + MDA	Reduce egg input into water	Egg input must fall about 90% before transmission falls substantially, because asexual multiplication in the snail amplifies the parasite
Health education	Reduce human water contact	—
Snail control (molluscicides, biological control by predators or competitors)	Fewer intermediate hosts; theoretically proportional reductions in transmission ; some call it the best single measure	Adult worms live for years , so snail control alone must continue for many years

- **Integrated campaigns** (treat people + separate people and their waste from water + control snails) are most promising.
- **Water, sanitation and hygiene (WASH) programmes** are back at the centre of control, with wider benefits.
- **Vaccine: none effective yet.**
- **Unlike malaria, lymphatic filariasis and onchocerciasis, global prevalence has not fallen, but schistosomiasis is eliminated in 10 countries: Iran, Japan, Lebanon, Malaysia, Martinique, Montserrat, Morocco, Thailand, Tunisia, Turkey.**
- **65th World Health Assembly, May 2012: resolution WHA65.21 called for intensified control and elimination campaigns — a shift from morbidity control to elimination.**

KEY POINTS: SCHISTOSOMIASIS

- ***S. haematobium* worms survive in the host for years to decades.** A careful travel and social history identifies exposure, links it to UG symptoms and guides testing.
- **PZQ in early-stage UG schistosomiasis can reverse the inflammatory lesions, including fibrosis, caused by the host response to eggs in tissue.**

- **Gold standard for active infection: eggs in urine, stool, or bladder or rectal biopsy. Serology and PCR are highly sensitive but may not distinguish active from resolved infection, and they are impractical in endemic regions.**

Concept — Schistosomal management follows the stage of the lesion. Egg-driven polypoid disease is reversible with PZQ, so treat medically first. Fibrotic stenosis is not reversible and needs surgery. Transmission is held up by snails and sanitation, so drugs alone reduce morbidity but cannot eliminate the disease.

Filariasis

- Filariae are vector-borne tissue nematodes. Human pathogens include the agents of **lymphatic filariasis (LF)**, *Onchocerca volvulus* and *Loa loa*.
- LF is caused by the mosquito-borne *Wuchereria bancrofti*, *Brugia malayi* and *Brugia timori*, spanning acute lymphatic inflammation to chronic lymphatic dilatation, with hydrocele, lymphoedema and elephantiasis.

Organisms

- **Transmission and development:**
 - A mosquito bite deposits **third-stage (infective) larvae**, which migrate to the **central lymphatics** and mature over **6–9 months**.
 - Adults (about 20–100 mm × 0.2 mm) live in the **afferent lymphatics**, especially of the **lower limbs** (inguinal, iliac, para-aortic); *W. bancrofti* alone also in the **male genitalia** (epididymis, spermatic cord, testis). Adult lifespan 5–7 years.
 - Females release microfilariae (about 200 × 10 µm).
 - **Periodicity:** microfilaraemia is **nocturnal (peaks at midnight)** in most areas, matching mosquito biting; **diurnal in parts of the Pacific**.
 - In the mosquito, microfilariae become infective larvae over 10–14 days.
- **Wolbachia:** an **obligate rickettsia-like endosymbiont** of *W. bancrofti* and *Brugia*, needed for embryogenesis. Doxycycline kills it, reducing microfilarial release and suppressing larval moulting.

Epidemiology

- About 68 million people have LF, a large fall since the early 2000s.

LYMPHATIC FILARIAE — DISTRIBUTION AND GENITAL INVOLVEMENT

Species	Share and distribution	Genital disease	Limb pattern	Circulating antigen test
<i>W. bancrofti</i>	> 90% of infections: sub-Saharan Africa, South and South-East Asia, western Pacific. Americas: only Haiti, Dominican Republic, Guyana, Brazil	Almost exclusively this species; very common	Whole limb	Yes (ELISA, ICT card)
<i>B. malayi</i>	Asia and Pacific islands (e.g. Indonesia, Philippines)	Uncommon	Leg below the knee	None (antibody dipstick available)
<i>B. timori</i>	South-eastern Indonesia only	Uncommon	Below the knee	None

- **The trap: genital filariasis means *W. bancrofti*;** Brugian infection spares the genitals and affects the limb below the knee.

- **Vectors:** *Anopheles* (rural Africa, Pacific), *Culex* (urban areas, especially India), *Aedes aegypti* (some Pacific islands), and others.
- **Transmission is inefficient**, so **obstructive lymphatic disease needs repeated infection over many years** and is seen mainly in **long-term residents**.
- **Prevalence rises through childhood to the third or fourth decade**, then plateaus as adult worms accumulate. **Lymphoedema and genital disease are rare before age 10**.
- **About one-third of the infected have overt disease**, more likely in India, Papua New Guinea and Africa, less in the Americas.

Pathology and Clinical Manifestations

- **Immune phases:**
 - **Early (larvae, young adults): pro-inflammatory, mixed Th1 and Th2;** filaria-specific IgE rises and eosinophils kill microfilariae.
 - **With microfilaraemia:** T-cell responses **down-regulated** by IL-10, IgG4 blocking antibody and antigen-specific suppressor T cells.
 - **Some stay uninfected despite long exposure;** whether protective immunity develops is unclear.
- **Damage is cumulative** (progressive scarring, lymphatic obstruction). **Treatment does not readily reverse it but can halt progression** in active infection.
- **LF is rarely fatal but causes the third-highest loss of disability-adjusted life years (DALYs) among parasitic infections.**
- **How lymphoedema develops:**
 - **Parasite factors dilate lymphatics** → **secondary bacterial infection and inflammation against dying worms** (and pro-inflammatory *Wolbachia*).
 - **Histology:** **granulomas around worms**, ± **tissue eosinophils** (Fig. 31.15).
 - **Vicious cycle:** acute attack → worse lymphoedema → more infection → worse lymphoedema.
 - **End result:** **obliterated, scarred lymphatics**, with **elephantiasis or hydrocele**.

FIG. 31.14 Microfilaria of *W. bancrofti* in peripheral blood. *Source:* CDC, Division of Parasitic Diseases and Malaria. **FIG. 31.15** Section of an adult *Brugia* worm in a lymph node. *Source:* CDC.

Subclinical Infection

- **Most infected people have few symptoms**, even with heavy microfilaraemia, yet almost all patent infections cause subclinical disease: **dilated lymphatics, scrotal lymphangiectasia, microscopic haematuria, proteinuria**. Eosinophilia is very common.

Acute Adenolymphangitis

- **Acute adenolymphangitis (ADL) is often the first manifestation:** fever, lymphadenitis, lymphangitis and oedema for days to a week.

FILARIAL VS BACTERIAL LYMPHANGITIS

Filarial (ADL)	Bacterial
Retrograde: spreads peripherally , away from the node	Ascending: spreads centrally

- **Genital lymphatics (*W. bancrofti*):** funiculitis, epididymitis, scrotal pain and tenderness; lymph scrotum — ruptured scrotal lymphatic vesicles → **whitish discharge** and **secondary bacterial infection**.
- **Dermatolymphangioadenitis (DLA):** fever, chills, myalgia, headache; oedematous inflammatory plaques, hyperpigmentation, vesicles and ulcers, often at a **skin injury**. Inflammation spreads **proximally**; thought to be **bacterial**.

Lymphoedema

- **Limb oedema is the commonest chronic manifestation: lower limb more than upper**, usually **asymmetric** (limb pattern by species in the table above).
- The skin may **exude serous fluid**. **The breast can be involved in women**.

Genitourinary Manifestations

- **Male genital disease** (frequency by species in the table above); female genital involvement is poorly characterised and appears uncommon.
- **Genital disease is not usually seen before the teenage years**.
- **Acute painful (usually unilateral) epididymitis or funiculitis** with fever and malaise lasts several days — **one of the commonest consequences** of bancroftian infection.

Funiculoepididymitis

- **Palpable, cord-like swelling with oedema**. Usually **self-limiting**, but **recurrence and chronic lymphoedema** are common.
- **Sterility and orchitis are rare, because the testis is usually spared**.
- **Often mistaken for malignancy** → unnecessary surgery, **including orchidectomy**.
- **A varicocele can complicate the inflammation**, adding pain and swelling.
- **Bacterial superinfection is rare but severe: exquisite pain, septic thrombophlebitis**.

Hydrocele

- **The commonest result of chronic male genital disease**; can be **very large** (Fig. 31.16).
- In endemic areas **filarial and non-filarial hydroceles are hard to distinguish**; **parasites are rarely found in the fluid**.
- **Suggesting LF: cord or epididymal nodules plus residence in or travel to an endemic area**; a **thick fibrous tunica**, especially with **cholesterol or calcium deposits**.
- **Painless** unless complicated by acute epididymitis or funiculitis. Scrotal skin may be **thickened, brawny, oozing lymph**.
- **Superinfection: rare in filarial hydrocele; common with elephantiasis and lymph scrotum**.

Scrotal and penile elephantiasis

- **Mild scrotal oedema** is common early or with hydrocele; **penile oedema is unusual**.
- **Massive scrotal or penile enlargement appears late**, mainly where care is poorly accessible.
- **Genital elephantiasis rarely has a cause other than LF**.

Chyluria

- **Damaged GU lymphatics leak lymph into urine** → **heavy fat and protein loss**, possible **malnutrition**. Rare.
- **Occurs earlier than genital elephantiasis**, is **usually intermittent**, and can **remit spontaneously**.

Tropical Pulmonary Eosinophilia

- **Tropical pulmonary eosinophilia (TPE): paroxysmal**, mainly **nocturnal cough and wheeze**, fever, lymphadenopathy, **high-grade eosinophilia, raised IgE**.
- **Cause: allergic response to microfilarial antigens; commonest in South and South-East Asia**.
- **Chest radiograph:** normal to diffuse reticulonodular shadowing. **Lung function:** restrictive (sometimes obstructive). **Biopsy:** eosinophilic interstitial pneumonitis.
- **Microfilaraemia is usually absent** — the trap: a negative blood smear does not exclude TPE.

FIG. 31.16 Huge hydrocele and scrotal elephantiasis.**Diagnosis**

- **Epidemiological context:** in endemic areas lymphoedema or male genital disease is **more likely LF**, once other causes of secondary oedema are excluded.
- **Differential of funiculoepididymitis:** TB, *S. haematobium*, gonorrhoea. **Non-filarial hydrocele is common everywhere**, but in endemic areas hydroceles are **more frequent and appear younger**.

DIAGNOSTIC TESTS FOR LYMPHATIC FILARIASIS

Test	Detail	Limitation
Adult worm on histology	Definitive	Insensitive ; worms lie deep in lymphatics
Ultrasound of lymphatics	" Filarial dance sign ": characteristic movement of live adults; ≥ 80% sensitive in some settings	Operator-dependent
Plain radiograph	Calcifications, suggestive in context	Non-specific
Giemsa-stained blood smear	Microfilariae. Time sample to local periodicity (usually midnight)	Positive in only a minority of infected
Circulating antigen (ELISA; immunochromatographic card test, ICT)	<i>W. bancrofti</i> only. Alere Filariasis Test Strip more sensitive in the field than older BinaxNOW ICT	None exists for <i>Brugia</i>
PCR (blood)	<i>W. bancrofti</i> and <i>B. malayi</i> ; very sensitive	Not widely available
Antibody	Traditionally poorly specific . IgG4 less cross-reactive, so more specific; species-specific antigens help; dipstick for <i>Brugia</i>	Cannot confirm active infection
Lymphoscintigraphy	Reliably shows lymphatic abnormality and extent	Does not distinguish LF from other lymphatic disease

- **"Burned-out" infection** (after treatment, or years after leaving an endemic area): **lasting lymphoedema or genital damage with negative microfilariae and antigen. Negative tests do not exclude LF**; these patients are **usually antibody positive**.

Treatment

- **Treat all patients with microfilaraemia, symptomatic or not**, because almost all have subclinical disease.

ANTIFILARIAL DRUGS — TARGETS, DOSES AND TOXICITY

Drug	Target	Regimen (LF unless stated)	Signature caution
Diethylcarbamazine (DEC)	Microfilaricidal; only modest activity against adults	Drug of choice for active LF: 2 mg/kg orally three times daily. 1-day course as effective as the traditional 12-day course. TPE: 2–3 weeks. In the United States available only through the CDC	Contraindicated where onchocerciasis or <i>Loa loa</i> may coexist (West and Central Africa) unless excluded. Reactions: fever, chills, arthralgia, myalgia, headache, nausea, vomiting; in heavy infection painful nodules, lymphadenitis, epididymitis from dying worms and Wolbachia
Albendazole	Microfilaricidal and macrofilaricidal	400 mg twice daily for 21 days	Few side effects in single-dose MDA regimens

Drug	Target	Regimen (LF unless stated)	Signature caution
Ivermectin	Mostly microfilaricidal	150–400 µg/kg once	Side effects similar to DEC; caution if <i>Loa loa</i> possible
Doxycycline	Anti- <i>Wolbachia</i> ; some macrofilaricidal activity; sterilises adults	200 mg daily for 4–8 weeks (6 weeks reasonable in the United States)	Not in pregnancy or young children; long course hard to deliver in resource-poor settings

- **What the table is for: most antilhelmintics kill microfilariae, not adults** — hence control needs **years of repeated dosing** until adults die of old age.
- **Doxycycline is the exception: it improves lymphoedema and hydrocele even without active infection**, suggesting benefit **beyond its anti-*Wolbachia* action**.
- **The trap: DEC with onchocerciasis or loiasis can cause severe reactions**, including blindness in onchocerciasis.
- **Chronic lymphoedema: control morbidity by preventing secondary bacterial infection, hygiene, elastic stockings, elevation, physiotherapy. Antiparasitics only if infection is active.**
- **Surgery:**
 - **Lymphatic-venous and nodal-venous anastomoses** have reduced leg swelling, and **reconstruction** has helped genital disease; **long-term results unknown**.
 - **Genital elephantiasis rarely suits surgery; lymphadenectomy can worsen lymphatic drainage.**
 - **Funiculoepididymitis: decompress or excise filarial nodules** to preserve testis and cord; if recurrent, painful, deforming or with vascular involvement → **more radical surgery**.
 - **Hydrocele:**
 - **Aspiration gives immediate relief** but recurs without medical and definitive surgical treatment.
 - **Hydrocelectomy for large or symptomatic hydroceles: excision of the intact sac is the procedure of choice; inversion with partial excision** is an alternative.
 - **Suture or excise leaking or dilated lymphatics.**
 - **Small, non-enlarging hydroceles need no surgery.**
 - **Scrotal or vulval reconstruction** with removal of redundant tissue helps selected patients.

Prevention and Control

- **Personal protection:** avoid mosquitoes; **long-lasting insecticide-treated bednets (LLINs)** are proven for control and elimination.
- **Clearing microfilariae from a community interrupts transmission**, because **the mosquito needs microfilaraemia to transmit**. But **drugs do not kill all adults**, so **intermittent dosing must continue for years** until adults die of senescence. This works for *W. bancrofti* but is **harder for *Brugia***, which has animal reservoirs.
- **MDA: single annual doses of albendazole plus ivermectin or DEC** (sustained microfilaricidal effect):
 - **Africa: albendazole + ivermectin** (DEC avoided because onchocerciasis and *Loa* are co-endemic).
 - **Elsewhere: albendazole + DEC.**
 - **MDA is the main reason global prevalence has fallen; about 19 million hydroceles prevented since 2000, halving LF hydroceles.**
- **Triple therapy (DEC + albendazole + ivermectin):** small studies show **deeper, longer microfilarial suppression** than dual therapy, **with no extra serious adverse effects**.

Onchocerciasis

- **River blindness**, caused by *O. volvulus* (vector in the comparison table).
 - **99% of cases in Africa**; foci in **Brazil, Venezuela and Yemen**.
 - **About 17 million infected**, about half the 2000 figure, thanks to **MDA**.
 - Transmission is **inefficient and focal**.
- **Adults live up to 15 years (mean 9–10) in subcutaneous nodules**; microfilariae migrate through **skin and eye**.
- **Clinical features: dermatitis and pruritus, keratitis, chorioretinitis → blindness from corneal scarring** after many years.
- **Urological relevance: late "hanging groin" or scrotal elephantiasis** from recurrent lymphadenitis and lost skin elasticity. **Histology:** atrophic, fibrotic inguinal nodes with subcutaneous oedema and fibrosis. Occasionally **massive inguinal lymphadenopathy**.
- **Diagnosis: skin snips** for microfilariae; **adult worms in nodules; microfilariae in the anterior chamber on slit lamp**. Serology is less developed than for LF.
- ***Onchocerca lupi*** (a dog parasite): a **rare human pathogen — six United States residents since 2013** (five children), all in the **south-west**, with nodules containing adult worms (spine, skin and soft tissue, extraocular muscle).
- **Treatment:**
 - **Ivermectin 150 µg/kg once, repeated every 6–12 months until asymptomatic**.
 - It acts mainly on microfilariae, but **repeated doses shorten adult lifespan by about half or more**.
 - **Adverse effects:** fever, rash, dizziness, pruritus, myalgia, arthralgia, lymphadenopathy, from **dying filariae and Wolbachia**.
 - **Caution if Loa co-infection is possible**.
 - **Then doxycycline 200 mg/day for 6 weeks: kills or shortens the life of most adult females and sterilises the rest**, clearing or greatly reducing microfilariae.
 - **DEC causes blindness and systemic toxicity** from inflammation against dying parasites.

Loiasis

- ***Loa loa*: Central and West Africa**, transmitted by *Chrysops* flies. **Adults migrate through subcutaneous tissue**.
- **Clinical features:**
 - **Most have asymptomatic eosinophilia**.
 - Some have urticaria, **migratory subcutaneous swellings** and **"eye worm"** (adult crossing the conjunctiva).
 - **Haematuria and proteinuria in about 30%; lymphadenitis and hydrocele rare**.
- **Treatment:**
 - **DEC 2–3 mg/kg orally three times daily for 14–21 days** is effective; multiple courses may be needed.
 - **Treatment reactions:** pruritus, arthralgia, migratory swellings, fever, eye worms, diarrhoea, **renal failure**.
 - **Encephalopathy risk with high microfilaraemia (> 2500–8000 microfilariae/mL), especially with ivermectin; pre-treatment apheresis may reduce it**.

- **Albendazole 200 mg twice daily for 3 weeks** has a **lower encephalopathy risk than DEC** and may be safer with high parasitaemia.

THE THREE FILARIAL INFECTIONS COMPARED

	Lymphatic filariasis	Onchocerciasis	Loiasis
Organism	<i>W. bancrofti</i> , <i>B. malayi</i> , <i>B. timori</i>	<i>O. volvulus</i>	<i>L. loa</i>
Vector	Mosquito (<i>Anopheles</i> , <i>Culex</i> , <i>Aedes</i>)	Simulium black fly	Chrysops fly
Adult site	Lymphatics (± male genitalia, <i>W. bancrofti</i>)	Subcutaneous nodules	Migrating subcutaneously
Microfilarial periodicity	Nocturnal (diurnal in parts of the Pacific)	In skin and eye	Diurnal
Wolbachia	Yes	Yes	No
Urological disease	Hydrocele, funiculoepididymitis, elephantiasis, chyluria	Hanging groin, scrotal elephantiasis	Haematuria and proteinuria (about 30%); rarely hydrocele
Drug of choice	DEC (Africa: ivermectin + albendazole in MDA)	Ivermectin → doxycycline	DEC (albendazole if high load)
Contraindicated	DEC where <i>Onchocerca</i> or <i>Loa</i> is co-endemic	DEC	Ivermectin (and DEC) with high microfilaraemia → encephalopathy

Concept — Filarial treatment depends on which filaria is present. Drugs kill microfilariae rather than adults, so the killing is what causes harm: DEC blinds in onchocerciasis, and DEC or ivermectin can cause encephalopathy in heavy *Loa* infection. *Wolbachia* is present in LF and *Onchocerca* but absent in *Loa*, which is why doxycycline is useful in the first two and not the third.

Other Non-filarial Genitourinary Parasites

Echinococcosis

- ***Echinococcus granulosus*, a cestode (tapeworm), causes cystic echinococcosis (hydatid disease).**
- **Transmission: ingested eggs in contaminated food or water, or contact with infected dogs. High prevalence in pastoral communities:** South America, the Mediterranean littoral, Eastern Europe, the Middle East, East Africa, Central Asia, China, Russia, Australia.
- **Cyst sites: usually liver, less often lung; kidney third** (see table).
- **Natural history: initially asymptomatic, growing 1–2 cm per year → pain or palpable mass. Hydatiduria and renal colic in a minority. Renal function usually preserved.**
- **Imaging: thick-walled, fluid-filled spherical cyst, often with a calcified wall; the appearance defines the stage, which guides management.**
- **Serology is adjunctive only** (sensitivity 60–90%).
- **Treatment:**
 - **Puncture, aspiration, injection, re-aspiration (PAIR)** suits liver cysts, not lung or kidney cysts, which have only **surgical resection or antiparasitic drugs**.
 - **Albendazole** (regimen in table) is the recommended medical treatment; **surgical excision** for some lesions because of size or location.
 - **Cyst rupture can cause anaphylaxis.**
 - **Perioperative PZQ + albendazole** may reduce **secondary seeding**.

Enterobiasis

- *Enterobius vermicularis* (pinworm) occurs **worldwide**, temperate and tropical.
- **Life cycle:** worms live in the **proximal colon** → migrate to **perianal skin to lay eggs**, **infective after 6 hours**. Spread **person to person** (faecal–oral, hands, fomites).
- **Clinical features:**
 - Usually **asymptomatic**; **perianal pruritus** can be severe.
 - **Ectopic migration in women** via **vagina, uterus and tubes into the peritoneum**: dead worms and eggs cause **granulomas and adhesions** → **vulval and cervical granulomas, salpingitis, oophoritis, tubo-ovarian abscess, appendicitis, peritonitis**.
 - **Men:** rare epididymal involvement and inguinal hernia.
- **Treatment:** single-dose albendazole or mebendazole (**highly effective**; doses in table); alternative ivermectin 200 µg/kg once. **Treat household and close contacts**; the **2-week repeat** covers **reinfection and autoinfection**.

Amoebiasis

- *Entamoeba histolytica*: a **protozoan** spread **faecal–orally**, commonest in the tropics.
- Most infected are **asymptomatic**; **about 10% develop disease in other organs, including the kidney**. Cutaneous amoebiasis causes **painful perianal and genital ulcers**.
- **Treatment:**
 - Tissue phase: tinidazole 2 g/day for 3–5 days or metronidazole 750 mg three times daily for 10 days.
 - Then a luminal agent: paromomycin 8–12 mg/kg three times daily for 7 days or iodoquinol 650 mg three times daily for 20 days.

Trichomoniasis

- *Trichomonas vaginalis*, a common **sexually transmitted protozoan**, is covered in **Chapter 29**.

OTHER NON-FILARIAL GU PARASITES — SUMMARY

Parasite	Type	Transmission	GU manifestation	Treatment
<i>E. granulosus</i>	Cestode	Eggs from dogs, food, water	Renal hydatid cyst (third site, about 2–3%); hydatiduria	Albendazole 400 mg twice daily for 1–6 months ± surgery; no PAIR for the kidney
<i>E. vermicularis</i>	Nematode	Person to person, faecal–oral	Ectopic female genital tract and pelvic granulomas; rarely epididymis	Albendazole 400 mg or mebendazole 100 mg once; repeat at 2 weeks; treat contacts
<i>E. histolytica</i>	Protozoan	Faecal–oral	Renal disease; painful genital and perianal ulcers	Tinidazole or metronidazole, then paromomycin or iodoquinol
<i>T. vaginalis</i>	Protozoan	Sexual	—	See Chapter 29

KEY POINTS: LYMPHATIC FILARIASIS

- LF is caused by three species of tissue nematode transmitted by mosquitoes.
- LF is endemic in much of the tropics but transmitted inefficiently. Clinically apparent disease is usually seen only in **long-term residents** of endemic countries.
- **Diagnosis:** blood smears, antibody and antigen assays, imaging and clinical assessment.

- **Treatment of active LF is DEC. Ivermectin and albendazole are also active, and all three are used in mass chemotherapy.**